Wed, Sep 9 at 4:54 AM

hey continue our fyp chat here

Yep 😭 We can continue from here.

The **latest FYP direction I have** is:

> **"A Multi-Modal Framework for Quantifying Narrator Reliability in ʿIlm al-Rijāl Using Semantic Sentiment Analysis and Graph Neural Networks"**

with the core architecture roughly:

Arabic biographical texts → NER / entity extraction → semantic sentiment/reliability signals → narrator knowledge graph → GNN → quantitative reliability score → isnād-chain evaluation

and your intended sources include **Tahdhīb al-Kamāl, Taqrīb al-Tahdhīb, Sanad**, etc.

We also recently had that existential moment of:

> *"Maybe I should've done Quranic Sentiment Analysis for Maqāṣid Derivation instead 😭"*

But my position was basically: **don't abandon the current FYP just because the alternative sounds intellectually cooler.** Your current one is much more aligned with Data Science + NLP + graph learning, provided we tighten the scope so it doesn't become an impossible digital-humanities project.

If you have the **latest proposal document**, though, send it here and I'll work from the actual latest version rather than relying on our previous reconstruction.

no i just want us resume to talk about what i should be doing for FYP2

Ahh yes 😭 — **FYP2 execution**, not re-litigating the topic.

Given where your FYP1 ended, I'd treat FYP2 as a **build → experiment → evaluate → write** semester.

What you should be doing now

1. Freeze the research scope

Don't keep adding cool ideas. Lock:

- Dataset / books you'll use
- What exactly "narrator reliability" means computationally
- Which NLP model(s)
- Knowledge-graph representation
- GNN architecture
- Final output/score
- Evaluation methodology

Your biggest danger is **scope creep**, not lack of technical difficulty.

2. Build the dataset pipeline first

Before touching the GNN, get:

```
Arabic biographical text → cleaned text → narrator entities → narrator attributes → structured records
```

You need an actual dataset you can inspect manually.

For example:

```
Plain text

Narrator
 ├── name / aliases
 ├── teachers
 ├── students
 ├── birth/death
 ├── location
 ├── al-jarh statements
 ├── al-ta'dil statements
 └── source/reference
```

This is probably your **highest-priority FYP2 task**.

3. Establish a non-GNN baseline

This is important academically.

Before claiming:

> "GNN quantifies narrator reliability"

you need something to compare against.

For example:

```
Traditional rule/weighted scoring → ML baseline → GNN
```

Then you can actually demonstrate whether the graph approach adds value.

4. Build the Knowledge Graph

Something like:

```
Plain text
```

```
[Narrator A]
      │
      ├── taught_by → [Narrator B]
      ├── student_of → [Narrator C]
      ├── praised_by → [Scholar X]
      └── criticised_by → [Scholar Y]
```

Then convert that graph into something PyTorch Geometric/DGL can consume.

5. Only THEN build the GNN

Start simple.

I would **not** jump straight into some fancy GATv2 + heterogeneous graph + multimodal transformer monstrosity. 😅

Start with something like:

```
Node features → GraphSAGE/GAT → reliability prediction
```

Get *one working model* first.

6. Evaluation is where your FYP becomes a thesis rather than a demo

You need to answer:

> Does the computational system actually reproduce / correlate with established rijāl assessments?

So construct a labelled evaluation set from authoritative classifications.

Then evaluate things like:

- Accuracy
- Precision
- Recall
- F1
- Correlation with classical rankings
- Ablation: text features vs graph features vs both

That last one is particularly valuable because it lets you demonstrate whether the **multi-modal** component actually matters.

If I were you, I'd order FYP2 like this

Phase	Priority
🔴 Dataset construction	**NOW**
🔴 Text preprocessing + NER	**NOW**
🔴 Reliability annotation/schema	**NOW**
🟠 Knowledge graph	Next
🟠 Baseline scoring model	Next
🟡 GNN	After baseline
🟡 Multimodal fusion	After GNN works
🟢 Evaluation + ablation	Later
🟢 Thesis/report	Throughout

And honestly, **don't start by coding the GNN**.

Your first FYP2 milestone should be:

> **"I have a clean, reproducible dataset of narrators with structured biographical/reliability information and a clearly defined ground-truth label."**

Once that exists, the rest becomes an engineering/research problem.

If you want, we can also reconstruct your **actual FYP2 roadmap from now → final presentation**, week by week, including what you should finish this week.

this is u said to me

Assuming your objectives stay fixed, I would **not touch the title or objectives unless your supervisor explicitly agrees**.

Instead, I'd focus on maximizing marks in the areas the examiners already identified.

Priority 1 (Highest Impact)

Fix the Research Design chapter

This directly addresses Lai's comment.

Right now, I suspect your methodology reads like:

> Here's the pipeline → here's the experiments.

FYP2 should read:

> Research Question → Hypothesis → Experiment → Metric → Result → Conclusion.

Create a table like:

Research Question	Experiment	Metric
Does semantic information improve narrator reliability estimation?	Graph-only vs Graph+BERT	Macro-F1
Does graph structure contribute meaningful information?	Semantic-only vs Graph-only	Macro-F1
Does multimodal integration outperform individual modalities?	Semantic vs Graph vs Multimodal	Macro-F1

This one change alone will make the thesis feel much more academic.

Priority 2

Lock your evaluation framework NOW

Tan's comment is easy to fix.

Before coding anything else, decide:

Primary Metric

- Macro-F1

Secondary Metrics

- Accuracy
- Cohen's Kappa
- Spearman Correlation
- MAE

And then never change them again.

One mistake many FYP students make is:

> Run experiment → metric bad → change metric.

Don't do that.

Write them into Chapter 3 and keep them fixed.

Priority 3

Literature comparison table

This is practically free marks.

Create a table like:

Study	Method	Task	Dataset	Metric	Result
Mahmoud et al.	NarratorsKG	Narrator Disambiguation	AR-Sanad	Accuracy	X
Narrator2Vec	Embeddings	Narrator Representation	Hadith corpus	F1	Y
Your work	Multimodal GNN	Reliability Quantification	Sanadset + Taqrib	Macro-F1	TBD

Then in Chapter 5:

> Our system achieved X compared to Y reported by Z.

Even if you don't beat them.

Lai specifically asked for this.

Priority 4

Fix narrator matching

If I were betting money on where your biggest performance gain comes from, it's here.

This statistic scares me:

> 3 exact matches, 1099 fuzzy matches.

That means your entire pipeline depends heavily on fuzzy matching.

I would spend serious time improving:

- kunya normalization
- ibn/bin normalization
- nisba handling
- teacher/student context
- tabaqa information

A 5% improvement here may help more than changing the GNN architecture.

Priority 5

Build a proper ablation study

You already have the foundations.

Run:

Experiment A

Graph-only

Experiment B

Semantic-only

Experiment C

Graph + Semantic

Experiment D

Graph + Semantic + Temporal

Experiment E

Graph + Semantic + Temporal + Geographic

Then show:

Model	Macro-F1
Graph	0.xx
Semantic	0.xx
Multimodal	0.xx

Examiners love ablation studies because they answer:

> Which component actually matters?

Priority 6

Revisit Objective 3 interpretation

Even if you don't change the objective itself.

You can still adjust how you write about it.

If Objective 3 currently evaluates against Hadith gradings:

don't present it as:

> "The system grades Hadith."

Present it as:

> "Narrator reliability estimates were evaluated through their ability to produce chain-level assessments consistent with established Hadith classifications."

Subtle difference.

Same objective.

Stronger narrative.

Addresses Lai's concern.

Priority 7

Improve visualizations

Tan practically told you what he wants.

Keep only:

Figure 1

System Architecture

Figure 2

Knowledge Graph Example

Figure 3

Narrator Reliability Distribution

Figure 4

Model Comparison Bar Chart

Figure 5

Confusion Matrix

Figure 6

Example Chain Assessment

Every figure should answer:

> Why does this figure exist?

If it doesn't answer a question, remove it.

Priority 8

Strengthen the Discussion chapter

This is where many FYPs lose marks.

Don't just write:

> Graph+BERT got 0.25.

Write:

> The results suggest that semantic biographical information contains stronger reliability signals than graph topology alone, indicating that narrator trustworthiness is primarily encoded in scholarly assessments rather than transmission-network position.

That's the kind of interpretation examiners want.

Priority 9

Prepare for Viva Questions Now

I can already predict some.

Why GNN?

Why not just BERT?

Why Macro-F1?

Why use Taqrib?

Why not train directly on Hadith grades?

Why use continuous scores?

Why not use a Transformer on the graph?

Why is graph-only performance weak?

Why is semantic information useful?

Start writing answers now.

Priority 10

Don't add new major components

This is the trap.

You are already tempted by:

- Graph Attention Networks
- Better Arabic LLMs
- Multi-sanad reasoning
- Matn analysis
- Geospatial verification
- Temporal verification

My advice:

Freeze the architecture by Week 3–4.

After that:

- run experiments
- write results
- write discussion

Not new features.

If I were planning your semester, I'd roughly allocate:

September

- Research design fixes
- Evaluation framework
- Literature comparison
- Narrator matching improvements

October

- Final model training

- Ablation studies

November

- Analysis
- Visualizations
- Chapter 5 & 6

December

- Thesis polishing
- Viva preparation

The biggest thing I'd tell September-2026 Alan is:

> You're no longer in the "build more AI" phase.
>
> You're in the "prove something scientifically" phase.

That mindset shift is probably worth more marks than any new model architecture.

YES. This is the exact thread we should pick up from. And I think the key conclusion remains:

> **FYP2 is no longer primarily about building the system. It's about experimentally proving that the system works and understanding why.**

So if we're asking **"what should I actually do next?"**, I'd narrow everything above into this:

Your immediate FYP2 checklist

1. Lock the experimental design

- Define the RQs.
- Define hypotheses.
- Lock Macro-F1 + secondary metrics.
- Define exactly what constitutes the ground truth.
- Define Graph / Semantic / Multimodal variants.

2. Audit the existing dataset

Especially that **3 exact vs 1,099 fuzzy narrator matches** issue.
Before training anything, we need to know how trustworthy those 1,099 matches actually are.

3. Build the baseline first

Something like:

```
Plain text

Classical / rule-based
        ↓
Semantic-only
        ↓
Graph-only
        ↓
Semantic + Graph
```

Then your FYP has a genuine experimental story.

4. Improve entity resolution

This deserves serious attention before model architecture tweaking.

5. Train the final models + ablations

```
Plain text

              ┌ Semantic
Input ────────┼ Graph
              └ Multimodal
                    ↓
               Reliability
```

Then determine **which information actually contributes**.

6. Analyse, don't merely report
The interesting thesis question isn't:

> "Our GNN obtained 0.xx F1."

It's:

> **"What information about a narrator actually allows computational models to distinguish levels of reliability?"**

That's where your discussion becomes genuinely interesting.

And I would make one change to our previous roadmap

I **wouldn't wait until October to start writing**.

From now on, every experiment should immediately produce:

Experiment → result → interpretation → thesis paragraph

That prevents the November/December nightmare where you have 40 notebooks and suddenly have to remember what the hell you did three months ago. 😂

So our working loop should be:

> **Design → Run → Record → Interpret → Write**

rather than:

> Build everything → panic → write thesis.

And honestly, Alan, I think your next move should be **to establish exactly what you already have from FYP1**.

If you have the latest **FYP1 report/proposal + current code/results**, send/upload them here. I can then turn this into a concrete **FYP2 execution plan based on your actual current state**, rather than us planning from memory.

cant u access this chat https://chatgpt.com/share/6aa0768d-5c68-83ec-b1f0-d1684b3a18b1

I **can access the share link itself**, and I can confirm it resolves to your **"Latest FYP Discussion"** ChatGPT conversation. ChatGPT

However, the shared ChatGPT page isn't exposing the actual conversation messages to my web reader—the page comes back with **0 readable lines**. So I can't reliably extract the discussion from that link.

The good news is: **I already have a substantial amount of the FYP context from our previous conversation**, including the latest title, methodology, examiner feedback, dataset/matching issue, and the FYP2 priorities we were discussing.

If you want me to recover the **entire detailed discussion** rather than rely on the context I have, the easiest route is to **upload/export the conversation as a file** here.

Then I can go through it and say:

> **"Okay, this is exactly where we left off. Here's what you should do today."**

That would be much more reliable than me trying to reconstruct it from memory.

lemme copy stuff
Honestly? **It is substantially better than the earlier version.** It now feels like an actual Data Science FYP rather than a grand idea that happens to contain some ML.

My current assessment is **~8/10 as a proposal/early implementation**, with a realistic path to becoming a very strong FYP. The biggest issue is no longer "is this technically interesting?" It clearly is. The issue is **whether the final experiments will demonstrate that the proposed multimodal system actually adds value, rather than merely being a sophisticated pipeline.**

What I like most

1. The project has a real research structure now.

You have a clean progression:

classical text → parsing → narrator matching → KG → semantic features → GNN → chain scoring → evaluation

and the three objectives map reasonably well onto that pipeline. That is exactly the sort of structure that makes a supervisor/examiner able to see what you are actually contributing.

Even better, you didn't just propose it—you already built substantial pieces.

You currently have:

- **3,797** parsed biographical entries
- **1,030 / 1,200** Hadith aligned
- **1,102 / 1,229** narrators matched
- a graph with **42,021 nodes and 147,329 edges**
- a functioning sanad → lookup → GNN → scoring demonstrator.

That is a **very respectable FYP1 implementation**.

2. You accidentally discovered some genuinely interesting research findings

This is probably the most promising part.

Your GNN results aren't good—which sounds bad—but scientifically they're useful.

Graph-only:

- MAE = 0.248
- R^2 = −0.093
- Spearman = 0.068
- Pearson = 0.097

Graph + BERT:

- MAE = 0.265
- R^2 = −0.151
- Spearman = 0.169
- Pearson = 0.245

So the graph topology by itself is basically **not predictive of narrator reliability**, while semantic information gives a modest signal.

That is actually a much more interesting result than:

> "We put a GNN on a Hadith graph and got 97% accuracy."

Your data is suggesting something philosophically important:

> **Being central or well-connected in the transmission network does not necessarily mean being trustworthy.**

That actually makes sense for ʿIlm al-Rijāl, because **reliability is an attribute of the narrator's critical assessment, not simply their network position**.

That can become a very nice discussion section.

3. The data-engineering part is stronger than before

The two-stage alignment approach is sensible:

TF-IDF candidate retrieval → RapidFuzz verification → quarantine

rather than trying to perform expensive pairwise comparisons over the entire corpus.

And you've added reproducibility mechanisms:

- seed 42
- SHA-256 identifiers
- quarantine records

That's exactly the kind of detail that makes an FYP feel like serious computational research rather than a notebook experiment.

But here is where I'm going to be critical.

There are **three important conceptual problems** that I would fix before you lock the methodology.

1. Your biggest problem: the GNN is currently learning the answer you gave it

Look at your node features.

The graph includes:

> **Ibn Hajar grade score**

as a feature.

But your target is also narrator reliability.

That creates a massive leakage problem.

Essentially:

input: Ibn Hajar reliability score
target: narrator reliability score

So the model can simply learn:

> "This narrator has a grade of 0.85 → predict ~0.85."

That makes the GNN experiment much weaker scientifically.

You partially recognise this issue through the ablation study, but **the graph-only baseline should absolutely not contain the ground-truth grade feature if the purpose is to test whether graph structure predicts reliability.**

I'd restructure the experiments around:

A. Structural-only

degree, chain position, frequency, etc.

B. Structural + temporal/geographical

age, tabaqa, geographical feasibility, transmission mode, etc.

C. Structural + semantic

BERT / Jarḥ-Taʿdīl representation

D. Structural + semantic + temporal/geographical

full model

Then the question becomes:

> *What information does each modality contribute?*

That is much stronger.

---

2. The "Semantic Sentiment Analysis" terminology is shaky

This is probably the part I would be most nervous about defending in a viva.

You call the main contribution:

> **Semantic Sentiment Quantisation**

But what you're actually doing is closer to:

semantic ordinal regression / reliability-score regression over technical scholarly evaluations.

"Sentiment analysis" normally suggests something like positive/negative affect or opinion polarity.

Your task is much more interesting than ordinary sentiment analysis because:

> **Thiqa isn't simply "positive sentiment."**

The phrase:

> *laysa bi-thiqa*

contains a positive lexical item while expressing negative evaluation.

Your proposal correctly recognises this contextual problem.

But I'd seriously consider eventually framing this as something like:

> **Contextual Semantic Reliability Quantisation**

or

> **Semantic Reliability Regression**

rather than relying so heavily on the term "sentiment".

You can still explain that sentiment-analysis techniques are used, but the **research problem itself isn't sentiment analysis**.

That distinction could make your thesis academically cleaner.

---

3. The chain-level multiplication model needs more mathematical justification

This is the second big conceptual vulnerability.

You use:

```math
S_{chain}=\prod_i w_i
```

and then add edge modifiers.

The problem is intuitive:

Suppose every narrator is 0.85.

For 5 narrators:

```math
0.85^5 = 0.444
```

For 10:

```math
0.85^{10} = 0.197
```

So simply making a sanad longer causes the reliability score to collapse—even though all narrators have exactly the same reliability.

That's a **chain-length bias**.

And classical Hadith grading doesn't work as simply as:

> reliable narrator × reliable narrator × reliable narrator...

Your proposal itself says the original model also leaned toward a weakest-link function.

I think you should experimentally compare at least:

```math
R_{product}=\prod w_i
```

```math
R_{mean}=\frac{1}{n}\sum w_i
```

```math
R_{min}=\min(w_i)
```

and potentially a geometric/log aggregation that reduces length dependence.

Then **let the data decide which aggregation best predicts the reference grades**.

That becomes an actual research question rather than you arbitrarily deciding the mathematics beforehand.

Another thing I'd change: your ground truth

This sentence is potentially dangerous:

> "ground-truth Hadith classifications" from the Kutub as-Sitta.

The **collections themselves aren't all equivalent sources of uniform "ground-truth labels."**

Especially because your HuggingFace dataset appears to have a `Grade` field such as:

`"Sahih (Darussalam)"`.

Your examiner could ask:

> "Why are these grades ground truth?"

You need to be very precise about this.

I'd call them something like:

reference classifications

or

dataset-provided reference grades

unless you establish exactly who graded them, under what methodology, and how those labels correspond to your target taxonomy.

This is particularly important because your model is supposed to be grounded in **classical ʿIlm al-Rijāl**, while the evaluation label might be a modern editorial grading.

Those are not automatically equivalent.

Your narrator matching result is actually a huge warning

Only **3 out of 1,102** matches were exact. Almost everything required fuzzy matching.

That's not necessarily a disaster. In fact, it exposes a genuinely difficult Arabic entity-linking problem.

But an examiner can immediately ask:

> "How do you know the 80% fuzzy match is the *correct narrator*?"

That's more serious than the NER problem.

You could theoretically get:

Muhammad ibn X

matched to

Muhammad ibn Y

because their strings look similar.

So I would make your identity-resolution component increasingly **evidence-based**, not just string-based.

You already have the perfect ingredients:

name similarity + kunya + nisba + teacher/student neighbourhood + tabaqa + dates + geography

That should become your disambiguation logic.

Then you have a genuinely defensible research contribution:

> identity isn't resolved merely from lexical similarity; it is resolved through **multimodal historical evidence**.

That is much cooler.

And one thing I really like about your FYP

You have now created an interesting distinction between:

"Can the machine reproduce the classical judgement?"

and

"Can the machine detect structural evidence that supports or contradicts that judgement?"

That gives you a much richer framework.

For example:

Classical assessment:
Thiqa = 0.85

Graph evidence:
normal transmission neighbourhood

Temporal evidence:
student lived during teacher's lifetime

Geographic evidence:
possible co-location

Semantic evidence:
multiple corroborating taʿdīl statements

→ **high-confidence narrator profile**

versus:

Classical assessment:
Thiqa

but:

student/teacher temporal conflict
+
geographical impossibility
+
ambiguous transmission

→ **flag for scholarly review**

That is where your project becomes genuinely compelling.

Not:

> "AI decides whether a Hadith is authentic."

but:

> **"AI integrates textual, relational, temporal and geographic evidence to assist scholars in evaluating transmission reliability."**

Your proposal already explicitly preserves the human scholar's epistemological authority, which is the correct direction.

My scorecard right now

Dimension	My assessment
Research idea	9/10
Novelty	8.5/10
Data Science relevance	9/10
Engineering complexity	9/10
Methodological coherence	7.5/10
Current empirical evidence	7/10
Risk of methodological criticism	High
FYP feasibility	8/10
Potential final thesis quality	9/10

The funny thing is that **your current weak results make me more optimistic, not less**.

You've reached the stage where the system is telling you something about the problem. The graph isn't magically solving reliability. Semantic information appears more useful. Entity resolution is a much harder bottleneck than expected. Alignment has measurable failure modes. Those are actual research findings.

What I'd do for FYP2 is **stop adding fancy components indiscriminately** and turn the project into a controlled multimodal experiment:

> **What does each modality contribute, and does their combination improve reliability prediction compared with simpler baselines?**

That question can carry the entire thesis.

And honestly, Alan, this is now looking much more like something you can defend as **a proper Data Science research project with Islamic-studies grounding**, rather than merely "I made an AI Hadith checker."

Yep — **this is the context I wanted.** Now we're properly synced. 😭

And I think our FYP2 strategy should be built around one central principle:

> **Do not make the system bigger. Make the evidence stronger.**

Your FYP1 already demonstrated that you can build the pipeline. FYP2 needs to demonstrate that the pipeline produces a **scientifically defensible result**.

So, what should you do *right now*?

I'd make your immediate sequence:

① Audit the current data → ② fix leakage → ③ fix narrator matching → ④ establish baselines → ⑤ run controlled ablations → ⑥ analyse → ⑦ write.

Not GATv2. Not another LLM. Not matn analysis. 😄

The first thing I want from you

If you have the **latest FYP1 report/proposal and/or your current code/results**, upload them here. I want to inspect the actual methodology and implementation before we start changing anything.

Upload the latest FYP document and, ideally, the relevant current experiment/code files too. Then I can map:

> **what you already have → what is scientifically wrong → what needs fixing → exact FYP2 experiments → thesis chapters**

From there, we can turn this into a concrete **September → final viva roadmap**, rather than continuing to talk abstractly.

And I especially want to inspect the **current feature construction + target construction**, because the potential **Ibn Hajar grade leakage** is the first thing I'd verify before you train another model.

Fri, Sep 11 at 7:54 AM

Ok I think I need you to summarise back what am I actually trying to do with my FYP and what I wanna achieve

Yes. Let me strip away all the implementation details and describe **what your FYP is actually trying to accomplish**, because I think this is important for how you approach FYP2.

Your FYP in one sentence

You're trying to build a **data-driven framework that estimates the reliability of hadith narrators by combining information from classical ʿilm al-rijāl texts with the structure of their transmission networks**, and then use those narrator-level estimates to help evaluate isnād chains.

Your title captures this:

> **A Multi-Modal Framework for Quantifying Narrator Reliability in ʿIlm ar-Rijāl Using Semantic Sentiment Analysis and Graph Neural Networks**

But underneath the technical terminology, your actual research question is:

> **Can computational models integrate textual and relational evidence about narrators to produce a meaningful quantitative representation of narrator reliability?**

---

1. The problem you're trying to solve

Classical ʿilm al-rijāl contains enormous amounts of information about narrators.

For example, a scholar might say that a narrator is:

- trustworthy — **ثقة**
- truthful/reliable — **صدوق**
- weak — **ضعيف**
- abandoned — **متروك**
- liar — **كذاب**

But this information exists primarily as **unstructured Arabic scholarly text**.

At the same time, narrators aren't isolated individuals.

They form a huge transmission network:

```
Plain text

Malik
  ↓
```

```
Ibn Wahb
  ↓
Yahya ibn Yahya
  ↓
...
```

So you have **two fundamentally different kinds of evidence**:

Semantic/textual evidence

> What did the rijāl scholars say about this narrator?

Relational/graph evidence

> Who transmitted from whom, and what does the narrator's position in the transmission network look like?

Your idea is:

> **Can we computationally combine these modalities?**

That's the heart of the project.

---

2. What “quantifying reliability” actually means

You're **not** trying to replace the classical categories with an arbitrary AI score.

Rather, you're trying to transform qualitative scholarly information into a **computationally usable representation**.

Conceptually:

```
ثقة جداً
    ↓
high reliability

صدوق
    ↓
moderately high reliability

ضعيف
    ↓
low reliability
```

You want a continuous representation that allows computational comparison.

Something conceptually like:

```
Narrator A → 0.92
Narrator B → 0.78
Narrator C → 0.41
Narrator D → 0.08
```

The exact mapping and mathematical treatment need to be justified experimentally.

That number isn't supposed to mean:

> “Allah has revealed that this narrator is 92% reliable.”

😂

It means:

> **“Given the defined computational representation of the available evidence, this narrator occupies this position on the model's reliability scale.”**

That distinction is extremely important for your thesis and eventual scholarly positioning.

---

3. Where NLP comes in

Your first major challenge is turning the Arabic texts into structured information.

Essentially:

```
Classical Arabic rijāl text
          ↓
     NLP processing
          ↓
Narrator identification
          ↓
Reliability-related statements
          ↓
Semantic representation
```

You're particularly interested in **contextual meaning**.

This is why you ended up using BERT/AraBERT-type models rather than simple keyword counting.

Because:

ثقة

doesn't automatically mean positive reliability in every context.

For example:

ليس بثقة

contains the word **ثقة**, but the actual judgement is negative.

So your system needs to understand **contextual semantic meaning**, rather than merely count positive and negative words.

This is one reason I previously suggested that "**semantic reliability analysis**" may ultimately describe your work better than conventional "sentiment analysis."

---

4. Where the Knowledge Graph comes in

Then you have the second modality.

You construct a graph representing the narrator ecosystem:

```
      Scholar
         |
         | assessed
         ↓
     Narrator A
      /     \
     ↓       ↓
Narrator B   Narrator C
     |
     ↓
Narrator D
```

Edges can represent relationships such as:

- reported from
- heard from
- student of

- teacher of
- etc.

This gives you information that **text alone doesn't capture well**.

A narrator isn't just:

> “a person with the sentence ثقة associated with them.”

They're a node in a historical transmission network.

---

5. Then comes the GNN

This is where the Data Science contribution becomes particularly important.

A normal NLP model asks:

> **What does the text about this narrator say?**

A graph neural network asks something closer to:

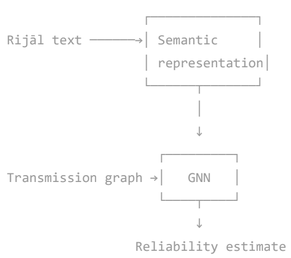
> **What can we learn about this narrator from their position and relationships within the transmission graph?**

Then your multimodal system asks:

> **What happens when we combine the two?**

Conceptually:

```
Plain text
```



And eventually:

```
Plain text
```

```
Semantic evidence
       +
Graph evidence
       ↓
Narrator reliability
       ↓
Isnād-level assessment
```

---

6. The really important research question

This is where I think your FYP became much more interesting than your original “AI grades hadith” idea.

You're **not merely asking**:

> “Can a GNN predict narrator reliability?”

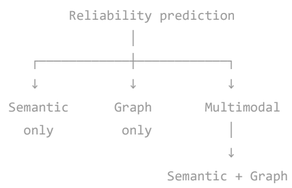
You're ultimately asking:

> **Which types of evidence actually contribute to computational estimation of narrator reliability?**

That's why the ablation study matters so much.

You want to compare something like:

```
Plain text
```



If graph-only performs badly but semantic information helps, **that's a result**.

If combining them improves further, **that's an even stronger result**.

If adding another modality doesn't help, **that's also a legitimate finding**.

You're discovering **what information matters**.

---

7. And then you move from narrator → isnād

This is the second layer.

Once you have computational reliability estimates for individual narrators:

```
Plain text
A = 0.95
B = 0.87
C = 0.73
D = 0.91
```

you can experiment with different ways of aggregating them into an isnād-level assessment.

This is where your current multiplication model comes in:

```
Plain text
R_chain = R_A × R_B × R_C × R_D
```

But, as we already identified, **you shouldn't assume multiplication is automatically correct**.

You want to investigate whether product, minimum, mean, geometric/log-based, etc. best corresponds to your reference classifications.

So this part of the FYP becomes:

> **Can narrator-level computational reliability estimates be meaningfully propagated to the chain level?**

---

8. What you're ultimately trying to achieve

So if I remove all the buzzwords, your FYP is trying to achieve **four things**.

Goal 1 — Digitise

Turn large quantities of classical rijāl scholarship into computationally usable structured information.

Goal 2 — Quantify

Convert qualitative narrator assessments into a reproducible numerical representation.

Goal 3 — Integrate

Determine whether **semantic evidence + transmission-network evidence** can provide a better representation of narrator reliability than either modality alone.

Goal 4 — Apply

Use those narrator-level estimates to computationally analyse isnād chains and compare the resulting assessments against established reference classifications.

---

9. What you DON'T want to claim

This is perhaps even more important.

Your FYP **is not trying to build:**

- ❌ An AI muḥaddith
- ❌ An automated fatwa system
- ❌ A machine that determines with certainty whether a hadith is authentic
- ❌ A replacement for classical ʿilm al-rijāl
- ❌ A new independent madhhab of hadith criticism 😄

Rather:

> **A computational decision-support and research framework for analysing narrator reliability using structured textual and relational evidence.**

The human scholar remains the final epistemic authority.

That's especially important because classical hadith criticism involves nuances that can't necessarily be reduced to your numerical representation.

---

10. And I think this is what you *really* want to achieve

There is a deeper motivation behind this project that I've noticed throughout our discussions.

You don't just want:

> **"I trained a GNN and got an F1 score."**

You want to demonstrate that **your Data Science skill can be applied to the Islamic intellectual tradition in a serious way.**

You're essentially building a bridge:

Classical Islamic scholarship

↕

NLP + knowledge graphs + machine learning

And ʿilm al-rijāl happens to be a fascinating domain for this because it already contains:

- structured relationships,
- historical entities,
- textual evaluations,
- temporal information,
- geographic information,
- transmission chains,
- and centuries of accumulated scholarly judgements.

That's almost tailor-made for computational modelling.

And if you execute this properly, **the FYP itself doesn't have to solve ʿilm al-rijāl**.

Its achievement can be much more defensible:

> **Demonstrate a reproducible computational framework for representing and analysing narrator reliability, and empirically determine how different modalities of historical evidence contribute to that task.**

That is a very respectable Data Science FYP.

And importantly, it gives you a foundation for the much larger things you're interested in later—computational ʿulūm al-ḥadīth, digital Islamic humanities, perhaps eventually your **Maqāṣid al-Qurʾān** idea.

So FYP2 isn't about making the dream bigger. It's about proving this first piece works.

So currently what are the issues you find and we can address on FYP2

Yes. Based on everything we've established so far, I'd separate the issues into **"must fix"**, **"should investigate"**, and **"nice to have."**

The good news is that I don't think your FYP has a fatal problem. Most of the weaknesses are exactly the sort of things FYP2 is supposed to resolve.

🔴 1. Data leakage in the GNN

This is the most important methodological issue.

If your input features include **Ibn Hajar's reliability/grade score**, while your target is also narrator reliability, then the model is being given information extremely close to the answer.

So:

> Input → "Ibn Hajar says this narrator is 0.85"
>
> Target → "Predict narrator reliability"

isn't a meaningful test of whether the **graph/semantic model** can estimate reliability.

FYP2 fix

Separate features into:

Structural

- degree
- transmission frequency
- neighbourhood characteristics
- position in chains

Temporal

- birth/death
- tabaqah
- temporal overlap

Geographical

- locations
- possible meeting/travel relationships

Semantic

- contextual representation of jarḥ/taʿdīl statements

And **exclude the target-derived grade from predictive features**.

Then test:

> Structural → Semantic → Structural+Semantic → Full multimodal.

That's a proper experiment.

🔴 2. Your narrator entity resolution is currently a major bottleneck

You had roughly:

> **3 exact matches vs ~1,099 fuzzy matches**

That's the number that would immediately make me nervous if I were an examiner.

The problem isn't fuzzy matching itself.

The problem is:

> **How do we know a fuzzy match refers to the same historical person?**

Arabic names are extremely tricky:

```
Plain text
```

```
محمد بن عبد الله
محمد بن عبدالله
محمد ابن عبد الله
أبو عبد الله محمد
محمد بن عبد الله البصري
محمد البصري
```

These can represent the same person—or completely different people.

FYP2 fix

Make entity resolution **evidence-based rather than string-based**.

Potential evidence:

- normalized name
- kunya
- nisbah
- father
- teachers
- students
- tabaqah
- birth/death
- location
- transmission neighbourhood

Then assign something like:

```
Plain text
```

```
Identity confidence
0.97 → very strong
0.84 → strong
0.61 → uncertain
0.32 → quarantine
```

And manually validate a statistically meaningful sample.

This could actually become one of the strongest parts of your methodology.

🔴 3. Your “sentiment analysis” terminology doesn't quite match the task

Your actual task isn't conventional sentiment analysis.

You're not asking:

> Is this sentence positive or negative?

You're asking:

> **What does this scholarly statement imply about a narrator's reliability?**

For example:

ثقة

and

ليس بثقة

cannot be handled by simple sentiment polarity.

FYP2 fix

You don't necessarily need to change the title now.

But internally, I'd conceptualise this as:

> **Contextual semantic reliability analysis / regression**

rather than ordinary sentiment analysis.

Then experimentally demonstrate that contextual embeddings capture reliability judgements better than lexical approaches.

🔴 4. Your reliability score mapping needs stronger justification

You need to defend:

> Why does **Thiqa = X** and **Ṣadūq = Y**?

If you're turning qualitative rijāl terminology into a continuous numerical scale, that mapping isn't self-evident.

There are several possible interpretations.

And different scholars don't necessarily use terminology identically.

FYP2 fix

Make the mapping an explicit **operationalisation**, not a claim that the numbers are metaphysically "the true reliability."

Then perform sensitivity analysis:

> Does changing the numerical mapping substantially change model performance?

If your results remain reasonably stable, that's actually useful evidence.

🔴 5. The chain aggregation model has length bias

Your current:

$$R_{chain} = \prod_i R_i$$

has a mathematical problem.

If every narrator has reliability 0.9:

```
Plain text
2 narrators → 0.81
5 narrators → 0.59
10 narrators → 0.35
```

The chain becomes worse merely because it contains more narrators.

That isn't necessarily what you want.

FYP2 fix

Experimentally compare:

- product
- minimum
- arithmetic mean
- geometric mean
- log-based aggregation
- possibly learned aggregation

Then evaluate which one best corresponds to your **reference classifications**.

This can become an actual research contribution rather than an arbitrary design choice.

🔴 6. Your “ground truth” needs to be extremely carefully defined

This is another thing I'd expect an examiner to attack.

If your Hadith labels come from a dataset containing things like:

> Sahih (Darussalam)

you can't automatically call that:

> **classical ground truth**

because that's a modern editorial classification.

Likewise, a narrator's assessment from Ibn Hajar isn't necessarily equivalent to the final grading of an individual hadith.

FYP2 fix

Use language such as:

> **reference classification**

unless you've established the provenance and methodology of the labels.

Then document:

Who produced the label?
What methodology?
What edition/source?
What does the label actually represent?

That makes your evaluation much more defensible.

🟠 7. Your graph may not contain enough information to predict reliability

Your preliminary results already suggest this:

Graph-only

- MAE ≈ 0.248
- R^2 ≈ −0.093
- Spearman ≈ 0.068
- Pearson ≈ 0.097

That's basically telling us:

> **Transmission-network structure alone doesn't explain narrator reliability very well.**

But this isn't necessarily a failure.

It could be one of your findings.

FYP2 question

Why?

Possibilities:

- graph topology genuinely isn't strongly associated with reliability;
- graph construction is noisy;
- entity resolution errors are contaminating the graph;
- your structural features are insufficient;
- labels aren't appropriate;
- model architecture isn't appropriate.

You need experiments that distinguish these explanations.

🔴 8. The multimodal model needs to actually demonstrate "multimodal value"

This is probably the **central scientific question of FYP2**.

Your thesis claims:

> **Multi-Modal Framework**

Therefore you need to prove that the modalities contribute something.

At minimum:

```
Plain text

Model 1: Semantic only
Model 2: Graph only
Model 3: Semantic + Graph
```

And ideally:

```
Plain text

Model 4: Semantic + Graph + Temporal
Model 5: Full multimodal
```

Then:

> **Does combining modalities actually improve prediction?**

If yes → great.

If no → that's still an important result.

🔴 9. You need proper baselines

A GNN performing at 0.XX doesn't mean much without comparison.

I'd want at least:

Baseline 1

Simple rule/score-based approach

Baseline 2

Traditional ML

e.g. Random Forest / Logistic Regression / XGBoost depending on target formulation.

Baseline 3

Semantic model

Baseline 4

GNN

Final

Multimodal model

Then your contribution becomes measurable.

🔴 10. Your evaluation needs to be tied directly to research questions

This is the issue we discussed from your examiner feedback.

Don't have:

> “We calculated F1 because F1 is commonly used.”

Instead:

RQ1

Does semantic information improve narrator reliability estimation?

→ Semantic vs structural

RQ2

Does transmission-network information contribute independently?

→ Graph vs semantic

RQ3

Does multimodal integration outperform individual modalities?

→ Semantic vs graph vs multimodal

RQ4

Can narrator-level estimates support chain-level classification?

→ chain aggregation experiments

Now every experiment has a **reason for existing**.

🟡 11. You need to investigate why the current model performs poorly

This is something I'd actually encourage rather than hide.

Your current result:

> Graph + BERT improves Pearson/Spearman somewhat but still has poor MAE/R^2.

Don't simply replace the model until you get a pretty number.

Do error analysis.

Take:

best predictions

worst predictions

false positives

false negatives

and ask:

> Why?

You might discover that the model fails on:

- ambiguous names
- rare narrators
- conflicting scholarly assessments
- sparse graph nodes
- long chains
- narrators with few textual descriptions

That turns a mediocre metric into an intellectually useful analysis.

🟡 12. Your dataset needs better documentation

By FYP2, someone should be able to reproduce:

```
Raw sources
 ↓
Cleaning
 ↓
NER
 ↓
Entity resolution
 ↓
Quarantine
 ↓
Feature extraction
 ↓
KG construction
 ↓
Train/validation/test split
 ↓
Model
 ↓
Evaluation
```

And you already have good foundations here:

- seed 42
- SHA-256 identifiers
- quarantine records

I'd build on that.

🟡 13. Train/test leakage needs careful checking

This is easy to overlook with graph data.

If related narrators or nearly identical texts appear across train and test, your performance could be artificially inflated.

You need to think carefully about:

- narrator-level splitting
- chain-level splitting
- temporal splitting
- duplicated textual statements
- connected graph leakage

This deserves an explicit methodology section.

🟡 14. The system needs to distinguish "prediction" from "scholarly evidence"

This is a conceptual issue I really want you to preserve.

Suppose:

> Classical assessment = Thiqa

but your computational evidence detects:

> temporal impossibility + questionable transmission relationship.

Your system shouldn't say:

> "**The narrator is actually unreliable.**"

It should say something closer to:

> "**The computational evidence conflicts with / warrants examination of the reference assessment.**"

That's much more academically responsible.

🟢 15. Explainability would significantly strengthen the final system

Not necessarily a huge new research component.

But ideally, when your system gives:

> Narrator reliability = 0.82

you can show:

```
Reliability estimate: 0.82

Semantic evidence: +0.18
Graph evidence:    +0.07
Temporal evidence: +0.04

Key statements:
• ثقة
• صدوق
• كان من أهل المدينة

Graph evidence:
• 14 transmission relationships
• 8 corroborating neighbours
```

Then your output becomes something a researcher can actually inspect.

So I'd rank the FYP2 problems like this

Priority	Problem
🔴 1	Data leakage
🔴 2	Narrator/entity matching reliability
🔴 3	Ground-truth/reference-label definition
🔴 4	Reliability-score operationalisation
🔴 5	Chain aggregation / length bias
🟠 6	Proving multimodal contribution
🟠 7	Proper baselines
🟠 8	Train/test leakage
🟠 9	Error analysis
🟡 10	Terminology (“sentiment”)
🟡 11	Explainability
🟡 12	Visualisation/documentation

The really important part

I **wouldn't treat all 12 as separate projects**.

They naturally collapse into **four FYP2 workstreams**:

A. Data validity

Entity resolution + labels + leakage + dataset documentation

B. Model validity

Baselines + leakage-free features + model architecture

C. Scientific evaluation

RQs + ablations + metrics + aggregation experiments

D. Scholarly interpretation

Error analysis + explainability + limitations + what the results actually mean

And **that should be your FYP2.**

The goal isn't to make the architecture increasingly complicated.

It's to get to the point where you can stand in your viva and confidently say:

> **"We tested whether textual semantic evidence and transmission-network structure independently contribute to narrator reliability estimation. Here is what each modality contributed, here is how we know, here are the failure cases, and here are the limitations of the computational representation."**

That is a much stronger FYP than simply saying **"my GNN achieved 87% accuracy."**

My biggest doubt right now is that I'm limited to Ibn Hajar and the fact I'm using pure scraping from Shamela. Could there be other ways to objectively get proper Rijali profiles currently whether Arabic or English sources

Yes — **this is probably the biggest upgrade I would consider for your FYP2.** Your concern is legitimate: if the system's "ground truth" is essentially **Ibn Ḥajar → scraped from Shamela → converted into a score**, then you're at risk of building a very sophisticated model whose target is ultimately just one scholar's compressed judgement.

The good news: **you absolutely don't have to do that.**

I searched what is currently available, and there are several better approaches.

First: I would change how you conceptualise the "rijāl profile"

Don't think:

> **Ibn Ḥajar profile → reliability score**

Think:

> **Historical narrator → multi-source evidence profile → computational reliability representation**

That is a *much* stronger FYP.

And it fits your actual research question better.

---

1. You have much richer classical sources than Ibn Ḥajar

For Sunni rijāl, you can construct profiles from multiple layers of the tradition.

Early / foundational sources

al-Bukhārī — al-Tārīkh al-Kabīr

Very useful because it is early and contains narrator identities, teachers, students, locations, etc.

Ibn Abī Ḥātim — al-Jarḥ wa al-Taʿdīl

This is particularly valuable for your project because it explicitly preserves statements of jarḥ and taʿdīl from earlier critics. The work isn't merely a list of Ibn Abī Ḥātim's own opinions; it transmits evaluations from numerous earlier authorities. Sunna +1

Then:

al-ʿIjlī — Maʿrifat al-Thiqāt

Ibn Ḥibbān — al-Thiqāt / al-Majrūḥīn

Ibn ʿAdī — al-Kāmil fī Ḍuʿafāʾ al-Rijāl

These give you independent evidence streams.

---

2. Then you have the major synthesis layer

This is where Ibn Ḥajar becomes useful — but **not as your sole ground truth**.

al-Mizzī — Tahdhīb al-Kamāl

This is probably the single most important source I'd investigate for your project.

It aggregates enormous amounts of earlier rijāl material around individual narrators.

Ibn Ḥajar's:

Tahdhīb al-Tahdhīb

is essentially a later condensation/synthesis with additional material.

And:

Taqrīb al-Tahdhīb

compresses the narrator into a very short final judgement.

A useful description of the hierarchy is that **Tahdhīb al-Kamāl is the more comprehensive source, Tahdhīb al-Tahdhīb is a condensation with Ibn Ḥajar's additions, and Taqrīb is an even more compressed assessment**. Islam.wiki

That means your current setup may actually be **starting at the wrong end**.

You're currently using:

> Taqrīb / Ibn Ḥajar → numerical label

when you could instead be doing:

> **Primary/earlier evaluations → evidence aggregation → Ibn Ḥajar as one later synthesis/reference**

That's much more intellectually defensible.

---

3. al-Dhahabī gives you another major independent stream

For example:

Mīzān al-Iʿtidāl

This is particularly valuable for **jarḥ / criticised narrators**.

It isn't simply a duplicate of Ibn Ḥajar.

And importantly, modern resources can expose these classical texts in structured form.

For example, Sunnah.com currently has individual narrator pages containing:

- Arabic name
- lineage
- kunya
- nisbah
- generation
- location
- teachers
- students
- critical appraisals
- source attribution
- original Arabic rijāl excerpts

For one example, its narrator record for Ziyād ibn ʿAmr shows appraisals attributed to Ibn Ḥibbān, al-Dhahabī and Ibn Ḥajar alongside teachers/students and the original Arabic material from **Tahdhīb al-Kamāl**. Sunnah.com

That's already much closer to the **structured narrator object** you want.

---

4. There are actually existing structured databases you should investigate

This is the part I think you'll find exciting.

Hadith-Viz

There's now a **Rijāl Database Browser** which, as of its 2026 update, reports:

> **73,003 searchable narrators**

and currently exposes five classical sources:

- al-Iṣābah
- Taqrīb al-Tahdhīb
- Tahdhīb al-Tahdhīb
- Mīzān al-Iʿtidāl
- Tajrīd Asmāʾ al-Ṣaḥābah

It also provides the original Arabic biographical excerpts and source attribution. Hadith Viz

This is extremely relevant to your FYP.

Because what you've been manually trying to construct from Shamela is conceptually much closer to what a dedicated Rijāl database is already attempting to provide.

I wouldn't automatically use it as your ground truth, though.

I'd investigate its **data provenance and underlying schema first**.

---

5. OpenITI is another important resource

This one is different.

OpenITI isn't primarily a narrator database. It's a **machine-actionable corpus of Arabic/Islamicate texts**.

Its current corpus has thousands of texts, with metadata and standardised machine-readable organisation. Open Islamicate... +1

And here's the interesting part:

OpenITI already contains large amounts of material originally sourced from Shamela, but in a more reproducible corpus infrastructure. Open Islamicate...

So your concern:

> "I'm just scraping Shamela."

doesn't necessarily mean your project is doomed.

The bigger issue is:

> **Are you treating the digitised text as a scholarly source with identifiable provenance?**

OpenITI can help you make the computational corpus side much more reproducible.

---

6. There's even a very interesting new structured corpus

I found something called the **Islamic Primary Source Corpus (IPSC)**, which currently advertises:

- ~26,923 narrators
- 20 classical biographical dictionaries
- six centuries of hadith scholarship
- structured narrator records
- attributed evaluations

and describes its Rijāl index as resolving narrator records across those classical sources. TheoAI

However: I would treat this as a potentially useful research lead, **not automatically as authoritative ground truth**.

It's a relatively new digital resource, so for an academic FYP I'd want to inspect:

- source editions
- entity-resolution methodology
- annotation methodology
- licensing
- whether the data are downloadable
- how disagreements between classical sources are represented

before depending on it.

---

7. This gives you a much better possible architecture

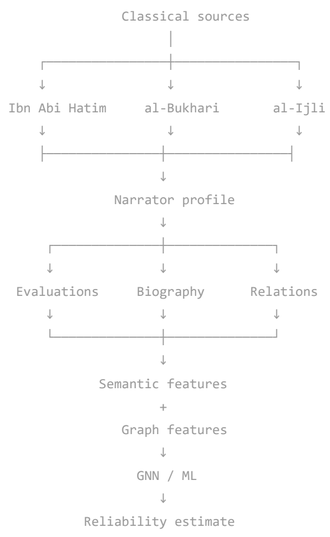
Instead of:

```
Plain text
```

```
Shamela
   ↓
Ibn Hajar
   ↓
Taqrib
   ↓
0.85
   ↓
GNN
```

I'd much rather see:

```
Plain text
```



Then **Ibn Ḥajar becomes one of the inputs/reference authorities rather than the definition of reality.**

That's a major improvement.

---

8. And here's the really cool possibility

You don't necessarily need to create a single “true” score from the classical sources.

In fact, **I think that would be methodologically worse.**

Suppose:

Scholar	Assessment
Yaḥyā b. Maʿīn	ثقة
Abū Ḥātim	صالح الحديث
al-Nasāʾī	ليس به بأس
Ibn Ḥibbān	ثقة
al-Dhahabī	صدوق
Ibn Ḥajar	صدوق

Your system can preserve:

```
Narrator X
│
├── Yahya b. Ma'in → Thiqa
├── Abu Hatim → Salih al-Hadith
├── al-Nasa'i → La Ba'sa Bihi
├── Ibn Hibban → Thiqa
├── al-Dhahabi → Saduq
└── Ibn Hajar → Saduq
```

That's actually valuable data in itself.

Then your model can learn from the **distribution and semantic content of the evaluations**, rather than pretending that one scholar's final label is absolute ground truth.

---

9. This also solves another problem you had

Remember our concern:

> "Why does Thiqa = 0.9?"

Exactly.

If you have multiple scholars, you can distinguish:

Classical evidence

> 5 critics assessed him positively, 1 criticised him.

from:

Computational representation

> Model's estimated reliability = 0.84.

That is much cleaner.

You're no longer saying:

> "Thiqa objectively equals 0.90."

You're saying:

> **"The model derives a continuous reliability representation from a corpus of attributed critical assessments."**

That's scientifically much easier to defend.

---

10. I would actually make disagreement a feature

This is where your FYP could become surprisingly sophisticated.

Imagine:

```
Narrator A
├── Critic 1: ثقة
├── Critic 2: ثقة
├── Critic 3: صدوق
├── Critic 4: ضعيف
└── Critic 5: متروك
```

Instead of destroying that information by choosing:

```
0.65
```

you preserve:

```
positive assessments
negative assessments
critic identity
date
school/context
exact quotation
source
```

Then the model has the opportunity to learn:

> **Reliability isn't merely a label. It is a historical body of scholarly evidence.**

That is much closer to what ʿilm al-rijāl actually is.

---

11. So should you abandon Shamela?

No.

I'd actually keep Shamela/OpenITI as a **primary textual corpus layer**.

The problem isn't:

> "Shamela is bad."

The problem is:

> **"One scraped source + one scholar's condensed judgement is too narrow to represent the rijāl tradition."**

OpenITI itself demonstrates that Shamela-derived digital texts can be incorporated into a machine-actionable scholarly corpus, while explicitly warning that digitised texts can differ in fidelity from print editions. Open Islamicate...

So your methodology could say something like:

> Classical biographical sources were computationally collected from machine-readable editions, with source, author, work and edition provenance retained for each evaluation.

That's considerably stronger than:

> "We scraped Shamela."

What I would recommend for YOUR FYP

I wouldn't suddenly try to ingest 50 books.

That would be classic **Alan Scope Creep™**. 😂

I'd aim for something like **5–8 carefully selected rijāl sources**, representing different chronological/critical layers.

For example:

Early:

- al-Bukhārī — *al-Tārīkh al-Kabīr*
- Ibn Abī Ḥātim — *al-Jarḥ wa al-Taʿdīl*

Intermediate:

- al-ʿIjlī — *Maʿrifat al-Thiqāt*
- Ibn Ḥibbān — *al-Thiqāt*
- Ibn ʿAdī — *al-Kāmil*

Later synthesis:

- al-Mizzī — *Tahdhīb al-Kamāl*
- al-Dhahabī — *Mīzān al-Iʿtidāl*
- Ibn Ḥajar — *Tahdhīb al-Tahdhīb / Taqrīb*

And then construct a **multi-source narrator profile**.

You don't even necessarily need every source for every narrator.

And this changes your FYP's scientific question in a good way

Instead of:

> **"Can AI reproduce Ibn Ḥajar's narrator grading?"**

you can investigate:

> **"Can a multimodal computational framework derive meaningful narrator-reliability representations from multi-source classical rijāl evidence and transmission-network structure?"**

That's **much more interesting**.

And then Ibn Ḥajar becomes one of your **reference authorities / comparative benchmarks**, rather than your entire epistemic foundation.

My strongest recommendation

Before you start scraping another 10 books, **let's first investigate what structured data already exists**—especially Hadith-Viz and the newer narrator corpora—and determine whether we can obtain **source-attributed, narrator-level records** rather than reinventing their entity-resolution work.

If we can get something like:

```
Plain text

narrator_id
canonical_name
aliases
teachers
students
birth/death
location
source
critic
assessment
original_text
```

then **that could radically improve FYP2**.

And importantly, we can still retain the original Arabic text and source provenance, so your model isn't becoming dependent on somebody else's opaque "0.87 reliability" labels.

Yooo omg I wished Hadith-Viz was around when I made FYP1 😭 I think currently that's the best dataset to use, coupled with my SanadSet dataset

HAHA yes 😭 **Hadith-Viz + your SanadSet is probably the most promising combination I've seen for rescuing/strengthening the FYP2 methodology without changing the fundamental project.**

I would actually be quite excited about this direction.

The important distinction is:

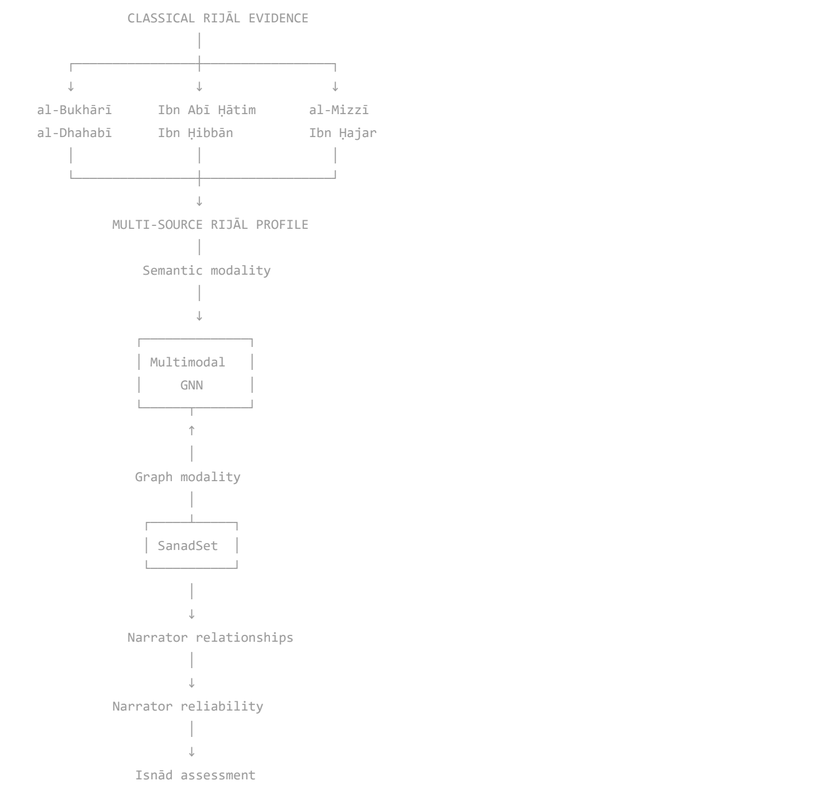
Hadith-Viz gives you the rijāl evidence layer

while

SanadSet gives you the transmission-network / hadith layer.

So you can finally make the “multi-modal” part of your FYP genuinely meaningful.

Plain text



And this fixes several problems simultaneously

Ibn Ḥajar limitation → multiple classical sources.

Shamela scraping limitation → structured/source-attributed database.

Weak graph-only results → much richer narrator profiles + better graph.

Ground-truth leakage → Ibn Ḥajar can become a **reference/benchmark**, rather than simply being fed into the model as the target.

“Sentiment analysis” problem → semantic representations can be derived from the actual attributed critical statements.

Entity matching problem → if Hadith-Viz already has canonical narrator identities, you potentially don't have to reinvent 1,100 fuzzy matches from scratch.

And there's a particularly beautiful experimental setup available:

Experiment	Rijāl	SanadSet graph
Baseline	Ibn Ḥajar / classical score	—
Semantic	Multi-source textual evidence	—
Graph	—	✓

Experiment	Rijāl	SanadSet graph
Multimodal	✓	✓

Then your central FYP2 question becomes:

> **Does integrating classical textual rijāl evidence with hadith transmission-network structure improve computational narrator-reliability estimation?**

That's *exactly* the question your title promises.

One caveat: don't blindly declare Hadith-Viz "the best dataset"

I'd call it **the best candidate currently available for your purpose**, rather than automatically adopting it.

Before we commit, I'd want to inspect its actual data/provenance:

1. **What exactly are the 73k narrator records?**
2. Are the evaluations individually attributed to specific critics?
3. Are the Arabic quotations preserved?
4. How are aliases resolved?
5. What editions are the texts based on?
6. Can the underlying data actually be accessed/downloaded?
7. What's the licensing situation?
8. Does it expose teachers/students in a machine-readable form?
9. How are contradictory assessments represented?
10. Most importantly: **does its narrator ID system align sufficiently with SanadSet?**

Because if the answer to #10 is yes, **holy shit, you've potentially solved one of your biggest FYP problems.**

You could use Hadith-Viz's canonical identity layer to connect:

rijāl profile ↔ SanadSet narrator node ↔ hadith isnād

rather than relying primarily on your old fuzzy matching pipeline.

And I wouldn't throw away your FYP1 work

That's important.

Your existing pipeline becomes your **original baseline / previous system**.

Then FYP2 can explicitly say:

> FYP1 established the initial computational framework using a single-source rijāl representation and preliminary entity matching. FYP2 extends this framework through multi-source rijāl evidence and improved narrator identity resolution.

That's actually a **very clean FYP progression**.

You're not admitting:

> "FYP1 was wrong."

You're saying:

> **"FYP1 established the prototype; FYP2 addresses the limitations identified through preliminary experiments."**

That's exactly how research is supposed to evolve.

And I think **Hadith-Viz + SanadSet + your existing NLP/GNN pipeline** gives us a genuinely compelling FYP2 without requiring you to reinvent the entire project.

If you can get me the **Hadith-Viz dataset/schema or export** you found, upload it here and I'll help you determine exactly how we should integrate it with your SanadSet.

Try investigate Hadith-Viz

Yooo, I dug much deeper into **Hadith-Viz**, and I need to correct/improve my earlier assessment.

This is considerably more interesting for your FYP2 than I initially thought. But there is also one major catch: **Hadith-Viz is not simply a downloadable "73k narrator dataset."** Its current architecture is much richer—and messier—in a way that actually maps very well onto your research problem.

1. First: the 73,003 figure is outdated

The January 2026 announcement said:

> 73,003 searchable narrators, from 5 classical sources.

But Hadith-Viz subsequently expanded dramatically.

As of the latest September 2026 information, the site says its **Rijāl database contains over 251,000 narrator biographies/entries**, and it holds **29 classical rijāl works containing 180,261 parsed biographical entries**. Hadith Viz +1

That distinction is important:

251k ≠ 251k uniquely resolved people.

The September update explicitly warns that the 180,261 entries represent *what the books contain under names*, and that a name appearing in multiple books does **not automatically mean it refers to the same person**. Homonyms are intentionally preserved. Hadith Viz

That's actually *excellent* for your FYP conceptually.

Why?

Because your current problem was:

> "How do I objectively build a narrator profile without simply trusting one Ibn Hajar entry?"

Hadith-Viz's answer is basically:

> **Keep the underlying scholarly entries and their provenance, then perform identity resolution separately.**

That's much closer to what you should be doing.

---

2. The classical-source coverage is REALLY good

Hadith-Viz currently says it integrates **14 classical biographical sources** into its narrator-verification layer:

Source	Author	What it contributes
Tahdhīb al-Tahdhīb	Ibn Ḥajar	biographies + assessments
Taqrīb al-Tahdhīb	Ibn Ḥajar	concise grades
al-Kāshif	al-Dhahabī	concise evaluations
Siyar Aʿlām al-Nubalāʾ	al-Dhahabī	extended biographies
Mīzān al-Iʿtidāl	al-Dhahabī	weak narrator evaluations
al-Iṣābah	Ibn Ḥajar	Companion identification
Usd al-Ghābah	Ibn al-Athīr	Companion biographies
al-Istīʿāb	Ibn ʿAbd al-Barr	Companion biographies
al-Thiqāt	Ibn Ḥibbān	trustworthy narrators
Tahdhīb al-Kamāl	al-Mizzī	comprehensive biographies
Tadhkirat al-Ḥuffāẓ	al-Dhahabī	hadith scholars
al-Jarḥ waʾl-Taʿdīl	Ibn Abī Ḥātim	criticism/validation
Lisān al-Mīzān	Ibn Ḥajar	supplement to *Mīzān*
Tārīkh al-Islām	al-Dhahabī	historical/narrator information

All 14 are sourced digitally through OpenITI according to Hadith-Viz's lineage documentation. Hadith Viz

And the broader corpus is actually **29 classical rijāl works**, totaling:

- **180,261 biographical entries**
- **173,288 with printed page references**
- **66,755 judgment statements**
- **400,838 transmitted statements from critics**
- **450,176 teacher/student relationships**
- **16,933 death years**

That's *extremely* relevant to your FYP. Hadith Viz

That last part is especially huge.

Your FYP wants:

> semantic evidence + relational/graph evidence → computational narrator reliability

Hadith-Viz potentially gives you **both sides of that equation**.

---

3. The really important thing: it preserves disagreement

This is probably the single biggest methodological advantage.

Hadith-Viz explicitly says:

> **"If the books differ, we preserve both statements rather than choosing for you."** Hadith Viz

That is *exactly* what I wanted your FYP2 to move toward.

Instead of:

```
Plain text
Narrator
   ↓
Ibn Hajar
   ↓
Thiqa
   ↓
0.90
```

you could have:

```
Plain text
                    ┌─ Ibn Abi Hatim → [assessment]
                    │
                    ├─ Ibn Hibban → [assessment]
Narrator ───────────┼─ al-Dhahabi → [assessment]
                    │
                    ├─ al-Mizzi → [assessment]
                    │
                    └─ Ibn Hajar → [assessment]
```

Then your model learns from the **distribution of classical evidence**, rather than pretending that one late scholar's condensed classification is the absolute ground truth.

That is a *much* more defensible research design.

---

4. And this solves one of your FYP1 problems

Remember your current issue:

> ~1,099 fuzzy matches out of ~1,102 narrator matches.

That's scary.

But Hadith-Viz itself doesn't pretend this problem magically disappears.

Its identity engine reportedly uses:

- name analysis
- teacher/student relationships
- death-year consistency
- Companion status
- cross-reference to classical biographies

rather than just string matching. Hadith Viz

And their September update is particularly revealing.

They explicitly rebuilt narrator identity and corrected cases where:

> a narrator's name merely contains another man's name

and stopped assigning inherited geography/reliability in such cases. They also deliberately leave ambiguous bare names unresolved rather than forcing an identity. Hadith Viz

That is exactly the philosophy you should adopt.

> **Uncertainty is better than a confidently wrong entity resolution.**

---

5. But here's the BIG catch

I would **not simply download Hadith-Viz and call it your ground-truth narrator dataset.**

Their own documentation warns that:

- source digitisation can contain OCR/transcription errors;
- their computational narrator extraction can make mistakes;
- identity matching is algorithmic;
- some entries remain ambiguous;
- some data is still being verified. Hadith Viz

And the full database page currently describes the broader database as containing entries with:

- incomplete biographies,
- conflicting evaluations,
- partial names,
- unresolved homonyms,
- lookup-only cross-references.

It even reports approximately **85% precision** for one calibrated sample of the AI-extracted full database tier. Hadith Viz

So:

Hadith-Viz = excellent *research corpus*

Hadith-Viz ≠ infallible *ground truth*

That's actually okay.

---

6. There's another fascinating thing: Hadith-Viz itself uses SanadSet

This is where your idea gets **really interesting**.

Hadith-Viz says its core hadith corpus is:

SanadSet 650K

with:

- 650,986 hadith records
- 960 books
- 926 extended collections
- 655,959 total after supplementary additions. Hadith Viz

And their narrator names originate from that SanadSet corpus, while the identity layer enriches those names against classical rijāl sources. Hadith Viz

So conceptually, you and Hadith-Viz are approaching the problem from almost opposite directions:

Your FYP1

```
Plain text
```

```
SanadSet
   ↓
Narrator extraction
   ↓
Fuzzy matching
   ↓
Rijāl
   ↓
GNN
```

Hadith-Viz

```
SanadSet
   ↓
Narrator extraction
   ↓
Identity engine
   ↓
Multiple classical Rijāl sources
   ↓
Narrator profiles
```

And this means your FYP2 could potentially become:

```
                SANADSET
                   │
                   ▼
        Transmission Graph
                   │
                   │
                   ▼
          ┌────────────────┐
          │   Narrator     │
          │    Entity      │
          │ Resolution     │
          └────────────────┘
                   │
          ┌────────┴────────┐
          ▼                 ▼
Classical Rijāl      Transmission
Evidence             Network
     │                    │
     ▼                    ▼
Semantic Features    Graph Features
     │                    │
     └────────┬───────────┘
              ▼
        Multimodal Model
              │
              ▼
   Computational Reliability
         Estimate
```

That's a much stronger thesis architecture than FYP1.

---

7. And Hadith-Viz's Rijāl data may be unusually useful for your semantic component

Look at the numbers again:

400,838 transmitted statements from critics.

That's potentially *far more interesting* than simply feeding:

ثقة / صدوق / ضعيف

into a classifier.

You could potentially distinguish:

Structured categorical evidence

```
Thiqa
Saduq
Da'if
Matruk
...
```

from

Raw scholarly linguistic evidence

```
ثقة
ليس به بأس
صدوق
صالح الحديث
ضعيف
متروك الحديث
كان يدلس
فيه نظر
ليس بالقوي
...
```

And then your Arabic NLP model can actually learn the **semantic structure of jarḥ wa-taʿdīl language**.

That makes the “Semantic Sentiment Analysis” part of your title much more meaningful.

---

8. The teacher/student data is also massive

Hadith-Viz reports:

450,176 teacher–student relationships. Hadith Viz

That could give you a second graph entirely.

You could distinguish:

Graph A — transmission graph

```
Hadith → narrator → narrator → narrator
```

from

Graph B — biographical relationship graph

```
Narrator
 ├── teacher
 ├── student
 ├── contemporary
 ├── region
 └── generation
```

And then combine them.

This would make your GNN much more meaningful than simply:

> “degree of narrator node”

which is one reason your graph-only experiment may currently be weak.

---

9. But don't accidentally use Hadith-Viz's own final grade as your feature

This is where I'd be very strict with your FYP.

Hadith-Viz displays narrator grades derived from **Ibn Hajar's *Taqrīb*** and *al-Kāshif*. Hadith Viz

If your target is:

> narrator reliability

then doing:

```
Plain text
Ibn Hajar grade
      ↓
feature
      ↓
model
      ↓
Ibn Hajar-like reliability
```

would basically recreate the label.

That's leakage.

Instead:

Better experiment

Model A — Structural

- degree
- in/out degree
- centrality
- transmission frequency
- chain position

↓

Model B — Semantic

- raw jarḥ/taʿdīl text
- critic identity
- assessment language
- contextual embeddings

↓

Model C — Temporal/Biographical

- death year
- generation
- teacher/student overlap
- geographical compatibility

↓

Model D — Multimodal

Everything above.

And **Ibn Hajar's final classification can be one evaluation/reference layer**, rather than blindly becoming an input feature.

---

10. Even better: critic-aware modeling

Here's where I think your FYP could become genuinely interesting academically.

Don't reduce:

> Ibn Abī Ḥātim: "X"

and

> Ibn Ḥajar: "Y"

into a single score.

Represent:

```
Plain text
Narrator X

Critic: Ibn Abi Hatim
Assessment: ...
Text embedding: [....]

Critic: Ibn Hibban
Assessment: ...
Text embedding: [....]

Critic: al-Dhahabi
Assessment: ...
Text embedding: [....]

Critic: Ibn Hajar
Assessment: ...
Text embedding: [....]
```

Then potentially:

```
Plain text
Narrator representation
=
f(
  scholarly assessments,
  critic identity,
  temporal context,
  network position,
  teacher/student structure
)
```

Now your model isn't merely predicting a label.

It's learning a **computational representation of historical rijāl evidence**.

That's much closer to the ambitious thing you originally wanted.

---

11. One thing I REALLY like about Hadith-Viz's methodology

Their latest update says they now show the **exact classical works carrying an entry under a narrator's name**, including:

- work
- edition
- entry number
- printed page

- exact name as printed

and currently search **29 classical works / 180,261 entries**. Hadith Viz

That is fantastic for reproducibility.

Your thesis could say something like:

> Each narrator profile is linked to source-attributed classical biographical evidence rather than treated as an untraceable aggregated label.

That's a much stronger scholarly statement.

---

12. API situation: this is important

Hadith-Viz announced in July 2026:

> **"API INCOMING!!"**

and described plans for downloading the full corpus, chains, gradings, and cross-references. Hadith Viz

However, the current academic-access page says that advanced research features—including the Rijāl database—require **academic access**, and asks researchers/students to contact them. Hadith Viz

So I **would not design your FYP around scraping the website**.

Instead:

Contact them as an academic researcher.

You're literally a UMS Data Science student doing a supervised FYP involving computational ʿilm al-rijāl.

That's exactly the kind of legitimate academic use they appear to be inviting.

---

13. And there's one thing we should NOT overlook

The current Hadith-Viz documentation says:

> **SanadSet 650K → Hadith-Viz**

So if you're thinking:

> "I'll use Hadith-Viz's SanadSet and my SanadSet"

we need to determine whether your SanadSet version is actually the same upstream dataset/version.

It may be.

If it is, **don't accidentally claim two independent datasets.**

Your contribution would instead be:

> **enriching the SanadSet transmission corpus with multi-source classical rijāl evidence.**

That's arguably *better*.

My current verdict

After investigating the actual current system, I'd upgrade my recommendation from:

> "Hadith-Viz looks like a good candidate."

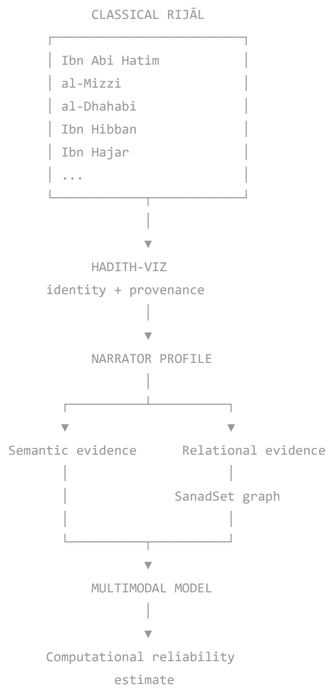
to:

Hadith-Viz is probably the strongest publicly accessible starting point I've found for your FYP2's rijāl layer.

But I would **not treat it as the dataset itself**.

I'd treat it as a **data lineage + enrichment ecosystem**:

Plain text



And **that** is much closer to the FYP you actually want to build.

The next thing I'd investigate

I think we should go one level deeper and figure out **whether Hadith-Viz's actual underlying Rijāl records/data are accessible**, what fields they contain, and—most importantly—**how its narrator IDs map to the SanadSet narrator names**.

That is the make-or-break question.

If we can establish a reliable mapping like:

Plain text

```
SanadSet narrator
        ↓
Hadith-Viz canonical identity
        ↓
14–29 classical sources
        ↓
individual critic statements
        ↓
teachers/students/death dates
```

then your FYP2 dataset architecture becomes **dramatically better than your FYP1 setup**.

Hadith-Viz Rijāl Database ↗
Hadith-Viz Sources & Data Lineage ↗
Hadith-Viz Methodology & Accuracy ↗
Hadith-Viz Academic Access ↗ 🔁

Yea I'm using the same 650k SanadSet

Ahhh — **that changes the situation significantly.** 😂

If your SanadSet is the **same 650k corpus that Hadith-Viz is built around**, then I would **not treat Hadith-Viz + SanadSet as two independent datasets**.

Instead, your FYP2 can be framed as **enriching the SanadSet transmission corpus with a substantially richer multi-source rijāl layer**.

Your FYP1

You essentially had:

SanadSet 650K → narrator extraction → fuzzy matching → Ibn Hajar/Taqrīb → GNN

The bottleneck was that you had to construct the rijāl side yourself, and the resulting ~1,099 fuzzy matches created uncertainty.

FYP2 could become

SanadSet 650K

→ canonical narrator entities
→ **Hadith-Viz multi-source rijāl enrichment**

→

- classical biographical evidence
- individual jarḥ/taʿdīl statements
- critic attribution
- teacher/student relationships
- death dates
- source provenance
- multiple scholarly evaluations
- semantic representations

PLUS

SanadSet transmission network

→ structural/graph features

→ **multimodal model**

→ computational narrator-reliability estimate.

And there's a beautiful methodological advantage here

You can now explicitly separate **data provenance** from **model prediction**.

For example:

> **SanadSet:** What did this narrator transmit, and with whom?
>
> **Hadith-Viz:** What do the classical rijāl sources say about this narrator?
>
> **Your model:** What quantitative representation emerges when these two evidence modalities are integrated?

That makes your contribution much cleaner.

You're no longer claiming:

> "I built an AI that determines whether Ibn Hajar is right."

Instead:

> **"I investigate whether classical textual rijāl evidence and transmission-network structure provide complementary information for computational narrator-reliability estimation."**

🔥 **That is a proper Data Science research question.**

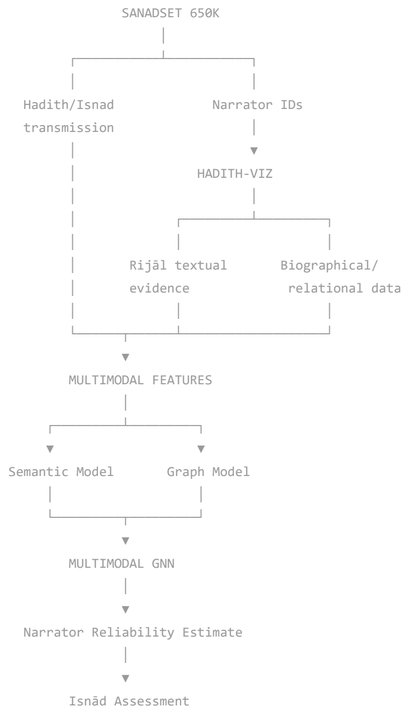
One BIG thing I would change from your FYP1

Don't make the 650K SanadSet itself the thing you need to "improve."

Your contribution is the **integration layer**.

Something like:

Plain text



And importantly, **don't count Hadith-Viz's 650K SanadSet as additional training data**. It is the same underlying corpus.

The novelty is the **rijāl enrichment and multimodal modeling**.

This also makes your FYP1 negative result more useful

Your graph-only model was weak.

That's actually potentially a great motivation:

> *Transmission topology alone may not contain sufficient information to estimate narrator reliability.*

Then FYP2 asks:

> **Does adding the semantic content of classical rijāl evaluations improve the model?**

That's a very clean experiment.

You can literally construct:

Experiment	Input
Baseline	classical/manual score
Graph-only	SanadSet graph
Semantic-only	Rijāl textual evidence
Graph + semantic	Both
Full multimodal	Graph + semantic + temporal/biographical

Then your central result isn't merely:

> "My GNN got 87%."

It's:

> **"Which evidence modality actually contributes information?"**

That's substantially more academically interesting.

And honestly, **the fact that Hadith-Viz uses the same SanadSet corpus is actually a blessing rather than a problem.** It means you're not trying to awkwardly merge two unrelated corpora with different hadith universes.

The thing I'd investigate next is **the actual Hadith-Viz narrator identity/data structure and whether you can obtain its IDs/exports/API**, because that determines whether we can replace your FYP1 fuzzy matching pipeline with a much more defensible entity-resolution layer.

If you have **your actual SanadSet schema/files or your FYP1 narrator-matching code**, uploading that would let me map out exactly how I'd redesign the pipeline for FYP2.

https://www.hadith-viz.com/rijal.html
Try navigating through to find narrators and see how quality they are

Yes — I tried to go beyond the landing-page claims and inspect how the actual narrator database behaves. **There is a very important distinction between the different Hadith-Viz Rijāl tiers.**

The first thing I discovered: don't use the 251k indiscriminately

The current Rijāl page explicitly divides the database into:

1. **Companions**
2. **With Sources**
3. **Full Database — 251,000+**

And Hadith-Viz itself warns that the Full Database contains incomplete records, conflicting evaluations, partial names and unresolved identities. More importantly, it gives its **AI-extracted Full Database only ~85% precision**, calibrated against a 100-entry stratified sample. Hadith Viz

That's actually a *very important* finding for your FYP.

I would rank the data quality roughly like this:

Tier	My assessment for your FYP
Companions	🟢 Very strong
With Sources	🟢🟢 **Best candidate**
Full Database / Classical confirmed	🟡 Useful research corpus
Full Database / homonym	🟠 Needs resolution
Full Database / lookup-only	🔴 Don't use as ground truth

Hadith-Viz itself distinguishes **Classical (confirmed)**, **Classical (homonym)** and **Classical (lookup-only)** records. The latter two are explicitly not equivalent to confirmed identities. Hadith Viz

---

And I found something REALLY important

The latest September 7 update says that when you open a narrator, Hadith-Viz now shows **which classical books actually contain an entry under that name**, including:

- book
- edition
- entry number
- printed page
- exact name as printed

It searches **29 classical works / 180,261 entries**. But—and this is crucial—the interface explicitly says that finding a name in a classical book **doesn't necessarily establish that the entry refers to the narrator you're viewing**. Hadith Viz

That is actually *excellent scholarly hygiene*.

They're essentially distinguishing:

> **name occurrence**

from

> **resolved person identity**

That's exactly the distinction your FYP1 fuzzy matching was struggling with.

The actual source base is impressive

The current documented 14-source verified layer contains:

- *Tahdhīb al-Tahdhīb*
- *Taqrīb al-Tahdhīb*
- *al-Kāshif*
- *Siyar Aʿlām al-Nubalāʾ*
- *Mīzān al-Iʿtidāl*
- *al-Iṣābah*
- *Usd al-Ghābah*
- *al-Istīʿāb*
- *al-Thiqāt*
- *Tahdhīb al-Kamāl*
- *Tadhkirat al-Ḥuffāẓ*
- *al-Jarḥ waʾl-Taʿdīl*
- *Lisān al-Mīzān*
- *Tārīkh al-Islām*

All 14 are sourced from OpenITI according to their documentation. Hadith Viz

And their broader 29-work corpus has **180,261 individual biographical entries**, with **173,288 (96.13%) carrying printed-page references**. Hadith Viz

That's much better provenance than:

> "I scraped Shamela and hope these names correspond."

😂

But here's what surprised me

The **grades themselves are still heavily centred around Ibn Hajar's Taqrīb**.

Hadith-Viz's documented grading system lists:

صحابي → ثقة ثبت → ثقة → صدوق → مقبول → ضعيف → متروك

and explicitly says the displayed grades are from **Ibn Hajar's *Taqrīb al-Tahdhīb***. Hadith Viz

So if you're thinking:

> "Great! I can just take Hadith-Viz's reliability grade as my new ground truth."

❌ **No.**

That would largely reproduce the exact methodological limitation we're trying to escape.

Instead, the **goldmine is underneath the grade**:

the individual classical source material and jarḥ/taʿdīl evidence.

And this is where Hadith-Viz becomes PERFECTLY aligned with your FYP

Their "With Sources" narrator profiles can expose things like:

- biographical excerpts
- reliability grades
- jarḥ/taʿdīl evaluations
- death year
- hadith count

- source attribution

and the site says profiles can draw from up to **14 classical sources**. Hadith Viz

So your data shouldn't ideally become:

```
Plain text

Narrator → 0.9
```

It should become something closer to:

```
Plain text

Narrator X
│
├── al-Jarḥ wa'l-Taʿdīl
│       └── critic assessment(s)
│
├── Tahdhīb al-Kamāl
│       └── biography + assessments
│
├── Tahdhīb al-Tahdhīb
│       └── biography + assessments
│
├── Mīzān al-Iʿtidāl
│       └── criticism
│
├── Taqrīb al-Tahdhīb
│       └── condensed grade
│
└── Siyar
        └── biography
```

Then your model determines how these evidence streams relate to reliability.

That is a much more interesting scientific problem.

The identity problem is also being handled much better than I expected

Hadith-Viz says its narrator identity engine uses multiple signals:

- name matching
- teacher/student relationships
- death-year consistency
- Companion status
- cross-referencing against classical sources

rather than merely matching strings. Hadith Viz

And their current system deliberately preserves homonyms rather than collapsing them.

They specifically give the example of multiple people named **Muḥammad b. ʿAbd Allāh** and say these are kept separate using information such as death year, nisba and kunya. Hadith Viz

That's huge for your FYP.

Because your FYP1 result:

> ~3 exact + ~1,099 fuzzy matches

was telling us:

string similarity alone is nowhere near sufficient.

Hadith-Viz essentially confirms that problem from the opposite direction.

One limitation I found that we need to respect

Hadith-Viz says:

> “no wrong data over no data.”

And the current system explicitly refuses to assign a grade when the grading cannot be verified. Hadith Viz

That's actually good.

For your dataset, I'd adopt the same principle:

Don't force every narrator into a class.

Instead:

```
HIGH CONFIDENCE
↓
usable for training

MEDIUM CONFIDENCE
↓
usable with weighting / sensitivity analysis

AMBIGUOUS
↓
evaluation or separate experiment

UNRESOLVED
↓
quarantine
```

That is far more defensible than trying to get 100% coverage.

And there's another really interesting data source

Hadith-Viz currently reports **450,176 teacher/student relationships** in its broader Rijāl corpus.

Combined with the SanadSet transmission graph, you potentially get:

Evidence layer

Classical Rijāl

```
critic → narrator → assessment
```

Transmission layer

SanadSet

```
narrator → narrator → narrator
```

Biographical relational layer

```
teacher → narrator → student
```

Temporal layer

```
birth/death → generation → possible contemporaneity
```

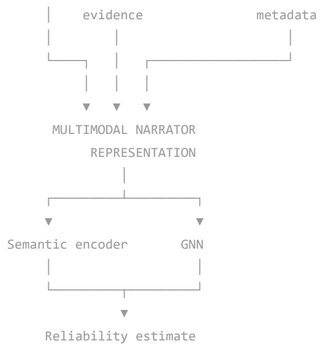
That's basically the multimodal representation you've been trying to build.

So I would change your FYP2 architecture slightly

Instead of:

```
SanadSet
   ↓
Fuzzy narrator matching
   ↓
Ibn Hajar
   ↓
GNN
```

I'd now aim for:



And critically:

Taqrīb grade should NOT simply be the input.

It can be:

- a reference label,
- one modality in a controlled experiment,
- or a comparative baseline.

But your model should have access to the **underlying scholarly evidence** if your goal is genuinely to quantify rijāl.

My quality verdict after actually examining the current system

Hadith-Viz is much better than "just another scraped Rijāl website."

I'd give it roughly:

Source provenance: 🟢🟢🟢🟢🟢
Classical-source breadth: 🟢🟢🟢🟢🟢
Narrator identity methodology: 🟢🟢🟢🟢
Transparency about uncertainty: 🟢🟢🟢🟢🟢
Raw Full Database reliability: 🟡
Verified/With Sources reliability: 🟢🟢🟢🟢
Suitability for your FYP: 🟢🟢🟢🟢🟢

The **251k number itself is NOT what impresses me anymore**.

What impresses me is:

> **29 classical works → source-level provenance → explicit homonym handling → teacher/student evidence → individual scholarly evaluations → same 650K SanadSet transmission universe.**

That's an unusually good fit for what you're trying to do.

And there's one more thing I think we should do before you commit: **sample actual narrator profiles systematically**—e.g. 10 famous narrators, 10 ordinary narrators, 10 ambiguous/common-name narrators, and 10 weak/criticized narrators—and score what fields are actually present and whether the source attributions are correct. That would tell us whether Hadith-Viz is genuinely usable as your FYP2 **data layer**, rather than merely looking impressive on paper. Hadith Viz +1

So why not limit the research to just full rijali profile? Since we just wanna showcase the main idea of quantifying

Yes — **for an FYP, I actually think that's the smarter scope.**

If your research objective is to **demonstrate the feasibility of quantifying narrator reliability**, you don't necessarily need to ingest all 251k+ Hadith-Viz entries or reconstruct the entire universe of rijāl literature.

The key is to define what you mean by **"full rijāl profile."**

I would scope it as:

> **A narrator is included only when Hadith-Viz provides a sufficiently complete, source-attributed profile for that narrator.**

For example:

```
Narrator X
│
├── Identity
│   ├── canonical name
│   ├── kunya
│   ├── nisba
│   ├── aliases
│   └── generation
│
├── Biographical evidence
│   ├── birth/death
│   ├── location
│   └── biography
│
├── Scholarly assessment
│   ├── critic
│   ├── assessment
│   ├── source
│   └── textual statement
│
├── Relationships
│   ├── teachers
│   └── students
│
└── Transmission
    └── SanadSet appearances
```

Then your dataset becomes something like:

SanadSet 650K → select narrators with sufficiently complete Hadith-Viz profiles → multimodal quantification.

That is much cleaner.

Why I prefer this over trying to cover everyone

Because your research question isn't:

> "Can I build the most comprehensive digital rijāl database?"

It's:

> **"Can multimodal computational methods meaningfully quantify narrator reliability?"**

Those are completely different projects.

Trying to resolve every obscure narrator introduces enormous amounts of **entity-resolution noise** without necessarily strengthening your proof of concept.

For an FYP, I'd rather have:

2,000 high-quality, well-resolved narrator profiles

than:

50,000 questionable profiles + 200,000 incomplete records.

You can explicitly state that **coverage was intentionally restricted to profiles meeting predefined completeness criteria**.

That's scientifically defensible.

In fact, it makes your experiment stronger

Suppose you establish:

> A "complete profile" must have at least:
- confirmed identity
- ≥1 source-attested rijāl assessment
- source attribution
- sufficient textual evidence
- identifiable transmission relationships

Then your experiment becomes:

Dataset construction

SanadSet 650K

↓

Extract narrator entities

↓

Match against Hadith-Viz

↓

Filter to complete profiles

↓

Final research cohort

↓

Train/test split

↓

Experiments

Model	Evidence
Baseline	Existing categorical/score baseline

Model	Evidence
Structural	SanadSet graph
Semantic	Rijāl assessment text
Biographical	temporal/geographical/profile features
Multimodal	all modalities

Then you ask:

> **Does combining the modalities improve quantitative narrator-reliability estimation?**

That's a *very* reasonable FYP.

There's an even deeper advantage

You don't have to claim:

> "This model quantifies all narrators."

Instead:

> "**This study evaluates the feasibility of computational narrator-reliability quantification using a curated cohort of narrators with sufficiently documented rijāl profiles.**"

That one sentence protects you from a **huge amount of scope creep**.

And it fits your original ambition perfectly.

I'd also avoid calling it "full" without defining it

Because "full rijāl profile" could be interpreted as:

> every book ever written about the narrator.

That's impossible for your FYP.

I'd call it something like:

Source-complete narrator profile

or

Multi-source rijāl profile

where *complete* means **complete enough according to your operational inclusion criteria**, not historically exhaustive.

For example:

> A narrator was considered eligible when Hadith-Viz provided a resolved identity and source-attributed rijāl evidence from one or more classical works, together with sufficient relational or biographical information for the intended feature modalities.

Then you can have:

Tier A — Multi-source complete
Best training cohort.

Tier B — Single-source documented
Potential secondary experiment.

Tier C — ambiguous/lookup-only/incomplete
Excluded.

And honestly, Alan, this is probably where I'd draw the FYP2 scope.

Don't spend six months trying to become **Hadith-Viz 2.0**. 😄

Use Hadith-Viz's enormous corpus as the **source layer**, use the same SanadSet 650K you're already working with as the **transmission layer**, select a defensible high-quality cohort, and spend your effort answering the actual scientific question:

> **Can semantic and relational evidence be transformed into a meaningful quantitative representation of narrator reliability, and does multimodal integration outperform individual modalities?**

If you can demonstrate that convincingly, **the proof-of-concept has succeeded**. The 251k-record expansion can then be explicitly identified as future work rather than something you need to solve for the FYP.

Niceee maybe u can try look up how complete the full rijali profiles in there

Yep — I checked the **actual numbers and development history**, and this gives us a much clearer answer.

The surprising part: the "full" 251k database is NOT 251k full profiles

Hadith-Viz currently reports **251,575 narrator biographies**, but the site explicitly says the Full Database includes people with limited data, conflicting evaluations, partial names, unresolved homonyms, and entries still being verified. It estimates this tier at about **85% precision** from a 100-entry stratified sample. Hadith Viz +1

So we should **not** interpret:

> 251,575 narrator biographies

as:

> 251,575 complete, research-ready rijāl profiles.

That's a crucial distinction.

But there's a much better number for what you're proposing

Hadith-Viz's own development history says:

> **3,140 narrator profiles with verified biographical entries across 14 classical sources + 1,162 verified Shamela matches.**

These were the curated/verified profiles before the huge 251k expansion. Hadith Viz

And their current documentation says a **verified narrator profile** can contain:

- biographies from up to 14 classical sources
- Ibn Hajar reliability grade
- al-Dhahabi reliability grade
- Jarḥ wa-Taʿdīl evaluations
- death year
- hadith count
- teacher/student relationships
- source citations. Hadith Viz +1

THAT is basically what we want.

There's an interesting evolution in their data

Their history makes the quality progression quite visible:

January 2026

They started with roughly:

73,003 searchable narrators

from five classical sources. Hadith Viz

Then they progressively built a verified layer.

February

They reported:

3,140 verified narrator profiles

across 14 classical sources, plus:

1,162 verified Shamela matches. Hadith Viz

They also explicitly said:

> **100% verified biographies**

after removing false matches and auditing the curated profiles. Hadith Viz

Then came May:

May 2026

They expanded the Rijāl browser from **3,140 → 251,000+**.

But the expansion was explicitly broader and less curated. They preserved homonyms and introduced categories such as:

- **Classical (confirmed)**
- **Classical (homonym)**
- **Classical (lookup-only)**

with the latter two carrying substantially more uncertainty. Hadith Viz +1

So your idea becomes even better

Instead of saying:

> "Let's use the full Hadith-Viz Rijāl database."

I'd formulate your dataset as:

Hadith-Viz verified multi-source narrator cohort

Essentially:

```
SanadSet 650K
      │
      ▼
Narrator mentions
      │
      ▼
Hadith-Viz identity matching
      │
      ▼
Verified narrator profiles
      │
      ├── Multi-source biographies
      ├── Jarḥ wa-Taʿdīl
      ├── Ibn Hajar
      ├── al-Dhahabi
      ├── death year
      ├── kunya / nisba
      ├── teacher/student
      └── source provenance
```

Then **exclude**:

```
Classical (homonym)
Classical (lookup-only)
unresolved
partial-only
no scholarly evaluation
```

unless you later want them for a secondary experiment.

And this may actually give you a VERY reasonable FYP dataset

Let's say after intersecting:

SanadSet 650K narrators

with

Hadith-Viz verified profiles

you get, hypothetically, 2,000–3,000 usable narrators.

That's **more than enough** for your proof-of-concept.

You don't need 251k.

You need enough diverse narrators to demonstrate:

- trustworthy
- generally trustworthy
- acceptable
- weak
- criticized
- different generations
- different network positions
- varying amounts of scholarly evidence

Then you can perform your multimodal experiment.

And Hadith-Viz actually gives you an excellent inclusion principle

Their philosophy is essentially:

No wrong data over no data.

Their August update says that if a grading cannot be verified, they simply **don't show a grade**, rather than borrowing one from another narrator with the same name. Hadith Viz

That's almost exactly the principle I'd adopt for your FYP dataset.

Your inclusion rule could be:

Include narrator if:

1. Identity is verified.
2. At least one classical source explicitly identifies the narrator.
3. At least one scholarly reliability assessment is available.
4. The profile can be confidently linked to a SanadSet narrator.
5. Required metadata for the experiment is present.

Everything else → **quarantine**.

One caveat I REALLY want you to notice

Hadith-Viz's own numbers are a little messy because they're talking about **different concepts**:

Number	What it actually represents
251,575	Current broad narrator-biography records
180,261	Entries across 29 classical rijāl works
3,140	Curated verified narrator profiles
1,162	Verified Shamela narrator matches

Number	What it actually represents
14	Classical sources in the main verified layer

So don't write in your thesis:

> "Hadith-Viz contains 251k complete narrator profiles."

That would be inaccurate.

I'd write something closer to:

> *Hadith-Viz provides a broad Rijāl database exceeding 251,000 biographical records, alongside a smaller curated set of verified narrator profiles derived from multiple classical biographical sources. This study restricts its experimental cohort to narrators satisfying predefined identity and source-completeness criteria.*

That's **much more defensible**. Hadith Viz +1

And there's one REALLY exciting possibility

Because Hadith-Viz says the 29 classical works contain **180,261 individual entries**, with **173,288 (96.13%) carrying printed page references**, we could potentially define *profile completeness* ourselves instead of trusting their 3,140 count. Hadith Viz

For example:

Profile completeness score

$$C_i = w_1(\text{identity}) + w_2(\text{source attribution}) + w_3(\text{jarḥ/taʿdīl}) + w_4(\text{biography}) + w_5(\text{temporal}) + w_6(\text{relational})$$

Then select:

> **$C_i \geq$ threshold**

for your research cohort.

That would actually make **your dataset construction itself reproducible**, rather than simply saying "Hadith-Viz said these are good."

And that's exactly the kind of Data Science methodology your supervisor can appreciate.

So yeah — I think your instinct is right. We don't need the entire 251k. We need a deliberately curated, source-complete subset of the 251k, linked back to the same SanadSet 650K.

That keeps the FYP manageable while preserving the central idea: **can we computationally quantify classical rijāl evidence when we have a genuinely rich narrator profile?**

Sat, Sep 12 at 6:36 AM

Finally last thing, should we add our own parsed Tarikh al-Kabir or its fine to leave it out?

I'd **leave *Tārīkh al-Kabīr* out of the primary dataset for now**.

Not because it's unimportant—it is actually a very valuable rijāl source—but because **Hadith-Viz already gives you a much cleaner, standardized multi-source evidence layer**, and adding your own parsed version creates another preprocessing/provenance problem that doesn't materially help prove your main hypothesis.

Why I'd leave it out

Your FYP2's core question is:

> **Does combining semantic rijāl evidence with transmission-network information improve computational narrator-reliability estimation?**

For that, your strongest design is:

Hadith-Viz verified multi-source profiles + SanadSet 650K

rather than:

Hadith-Viz + your own Shamela parsing + Tarīkh al-Kabīr + everything else.

Adding your parsed *Tārīkh al-Kabīr* would introduce:

- another text-extraction/OCR pipeline to validate;
- another entity-resolution problem;

- potentially duplicated evidence already represented through other sources;
- another source whose edition/provenance you need to document;
- more feature engineering;
- more opportunities for inconsistent preprocessing.

And none of that directly answers your research question.

But *Tārīkh al-Kabīr* is a fantastic future extension

I'd actually mention it explicitly in the thesis:

> *Future work may incorporate additional early rijāl sources, such as al-Bukhārī's Tārīkh al-Kabīr, to increase temporal and source diversity.*

That gives you a nice research-extension story.

There's also a methodological reason

Your current setup already has a pretty excellent source progression:

Early → middle → later criticism

with things like:

- al-Bukhārī's contemporaneous/early biographical tradition *if represented elsewhere*
- Ibn Abī Ḥātim's *al-Jarḥ wa'l-Taʿdīl*
- al-Mizzī's *Tahdhīb al-Kamāl*
- al-Dhahabī's works
- Ibn Ḥajar's *Tahdhīb / Taqrīb*

So you don't *need* to add every classical source to make the multimodal idea legitimate.

In fact, **source diversity is not the same thing as source quantity**.

I'd rather have a carefully documented set of ~10–14 sources with preserved attribution than throw 30+ books into the pipeline and then have no idea which evidence is driving the model.

One exception

If you've **already parsed *Tārīkh al-Kabīr*** and the data is clean, canonical, and already integrated into your FYP1 pipeline, then I wouldn't throw that work away.

We could use it as a **controlled supplementary modality/source**:

> Hadith-Viz-only vs Hadith-Viz + Tārīkh al-Kabīr

That could even become a small ablation:

Experiment	Rijāl evidence
R1	Hadith-Viz verified profiles
R2	Hadith-Viz + *Tārīkh al-Kabīr*
R3	R2 + SanadSet graph

But **I would not make R2 mandatory**.

My recommendation

Primary FYP2 dataset:

> **SanadSet 650K + Hadith-Viz source-complete narrator profiles**

Your existing *Tārīkh al-Kabīr* parser:

> Keep it as an optional extension / robustness experiment.

Do not spend FYP2 weeks trying to perfect it unless it comes almost for free.

Honestly, I think we've finally got a very clean scope:

> **SanadSet supplies the transmission world. Hadith-Viz supplies the classical rijāl evidence world. Your model investigates whether those worlds can be computationally integrated into a quantitative representation of narrator reliability.**

That's a much stronger FYP than trying to build the definitive digital *ʿilm al-rijāl* database.

Ok nevermind then i dont wanna add unnecessary burden. So i guess we can now compile a complete todo list for me to do for FYP2

Absolutely. I think we can now **freeze the scope** instead of continuously adding things.

The core FYP2 becomes:

> **Use the same 650K SanadSet transmission corpus, enrich its narrators with high-quality source-attributed Hadith-Viz rijāl profiles, and experimentally determine whether semantic, graph, temporal/biographical, and combined evidence can produce a meaningful computational representation of narrator reliability.**

The big rule for FYP2:

> **Do not build Hadith-Viz. Do not build a new rijāl database. Do not keep adding models. Build a clean experiment around the existing resources and answer the research questions properly.**

FYP2 MASTER TODO

PHASE 0 — Freeze the research design

0.1 Lock the research question

Your central RQ should be something close to:

> **RQ: To what extent can semantic and relational evidence from classical rijāl sources and hadith transmission networks be integrated to computationally quantify narrator reliability?**

Then divide it into measurable subquestions:

RQ1 — Semantic:
Does semantic information from classical rijāl assessments provide useful information for narrator-reliability estimation?

RQ2 — Structural:
Does transmission-network information provide useful information for narrator-reliability estimation?

RQ3 — Integration:
Does combining semantic and relational information outperform individual modalities?

RQ4 — Chain-level applicability:
Can narrator-level computational estimates be aggregated to provide useful assessments of isnād chains?

That gives you an extremely clean thesis structure.

0.2 Lock hypotheses

For example:

H1: Semantic rijāl features significantly improve narrator-reliability prediction over a simple baseline.

H2: Transmission-network features provide predictive information about narrator reliability.

H3: Multimodal semantic + graph modelling performs better than single-modality models.

H4: Narrator-level computational reliability estimates can support chain-level classification consistent with a reference classification.

Don't obsess over making every hypothesis statistically fancy. They need to map cleanly to experiments.

0.3 Lock the contribution

Your contribution is **not**:

"A new authoritative grading of narrators."

It is:

A multimodal computational framework for representing and quantifying narrator reliability from heterogeneous classical and relational evidence.

And explicitly:

The system is a **decision-support/research tool**, not a replacement for traditional hadith scholarship.

PHASE 1 — Rebuild the literature/research foundation

1.1 Update literature review

Build a proper comparison table covering:

- computational hadith research
- isnād analysis
- narrator-network analysis
- Arabic NLP for hadith
- knowledge graphs
- transformer/BERT-based Arabic semantic modelling
- GNNs on historical/religious networks
- computational reliability / credibility assessment
- existing digital rijāl resources

For each paper:

Paper	Data	Task	Method	Features	Metrics	Limitation	Your difference

This directly addresses your FYP1 feedback that the literature comparison wasn't sufficiently clear.

1.2 Document Hadith-Viz properly

Don't just write "we use Hadith-Viz."

Document:

- what Hadith-Viz is
- source lineage
- classical works represented
- narrator identity approach
- profile structure
- source attribution
- uncertainty categories
- limitations
- why you selected it

The broad Hadith-Viz corpus and the curated/source-supported narrator layer are not the same thing, so your methodology needs to distinguish them.

1.3 Document SanadSet

Make completely explicit:

- exact SanadSet version
- 650K records
- books represented
- fields available
- narrator representation
- hadith/reference labels
- how narrator chains were extracted
- any preprocessing already performed

Most importantly:

> **Hadith-Viz and your SanadSet use the same underlying 650K corpus, so they are complementary layers, not independent datasets.**

That sentence should probably appear somewhere in your methodology.

PHASE 2 — Dataset engineering

This is probably your **most important technical phase**.

2.1 Establish the canonical narrator table

Create something like:

```
Plain text
```

```
narrator_id
canonical_name
arabic_name
aliases
kunya
nisba
birth_year
death_year
generation/tabaqa
locations
identity_confidence
hadith_count
profile_status
```

One canonical row per resolved narrator identity.

2.2 Establish the narrator-source evidence table

Separate this from the narrator table.

```
Plain text
```

```
narrator_id
source_id
source_name
source_entry_id
printed_page
assessment_text
assessment_category
critic
assessment_date/period
```

This is very important.

Do not flatten all classical evidence into one column called `reliability`.

You want to retain:

> who said what, where, and in which source.

That makes your semantic experiment scientifically meaningful.

2.3 Establish the relationship table

Something like:

```
Plain text
```

```
source_narrator
target_narrator
relationship_type
source
confidence
```

Relationship types could include:

- narrated_from
- narrated_by
- teacher
- student

Keep provenance wherever possible.

2.4 Establish temporal/biographical features

Possible features:

```
birth_year
death_year
age/estimated lifespan
tabaqa
teacher_count
student_count
location_count
geographic overlap
possible contemporaneity
```

Don't create 70 random features.

Start with a **small theoretically motivated set**.

PHASE 3 — Define your “source-complete” cohort

This is one of the strongest ideas we've arrived at.

Don't use every Hadith-Viz record.

Define objective inclusion criteria.

For example:

Include

A narrator must have:

- resolved identity;
- confident linkage to SanadSet;
- at least one source-attributed classical rijāl record;
- at least one reliability-related scholarly assessment;
- enough information for the relevant experiment.

Exclude or quarantine

- unresolved identity;
- ambiguous homonym;
- lookup-only record;
- insufficient evidence;
- unreliable match;
- missing target/reference information.

Then record **why each narrator was excluded**.

3.1 Measure dataset attrition

This needs to become a table in your thesis:

```
Plain text

SanadSet narrator mentions
        ↓
unique narrator candidates
        ↓
matched to Hadith-Viz
        ↓
identity-confirmed
        ↓
source-supported
        ↓
reliability evidence available
        ↓
final experimental cohort
```

Report the number at every stage.

This turns your data-cleaning process into a reproducible scientific methodology rather than hidden preprocessing.

PHASE 4 — Entity resolution

This is where I want you to be especially careful because of your FYP1 experience.

4.1 Don't use pure fuzzy matching

Your old:

> exact + fuzzy name matching

should **not** be the final method.

Use a hierarchy of evidence:

Name

- **kunya**
- **nisba**
- **teacher/student relationships**
- **death year**
- **generation**
- **geography**
- **context**

4.2 Give every match a confidence score

For example:

```
Plain text
```

```
HIGH
MEDIUM
LOW
UNRESOLVED
```

Or numerical confidence if you want.

But importantly:

> **Low-confidence matches don't automatically enter training.**

4.3 Manually audit a sample

You don't need to manually inspect 20,000 people.

Take a statistically meaningful sample covering:

- common names
- rare names
- famous narrators
- weak narrators
- ambiguous names
- different generations

Record:

```
correct
incorrect
ambiguous
```

Then report your observed resolution quality.

This is much more valuable than simply saying "fuzzy matching was performed."

PHASE 5 — Define the target carefully

This is absolutely critical.

5.1 Decide what “reliability” means computationally

You need to determine whether your prediction target is:

Option A — categorical

```
Thiqa
Saduq
Maqbul
Da'if
Matrook
...
```

Option B — ordinal

```
higher reliability
       ↑
       |
       ↓
lower reliability
```

Option C — continuous

```
Plain text
0.00 ──────────── 1.00
```

My preference for the FYP is:

> **Use a classification/ordinal target for the primary experiment, while producing a continuous computational score as an interpretable output.**

That makes Macro-F1 usable as the primary metric while still supporting your "quantification" idea.

5.2 Do NOT blindly treat one scholar's grade as objective truth

Especially:

```
Plain text
Ibn Hajar grade
       ↓
0.8
       ↓
ground truth
```

That's too simplistic.

Instead, explicitly call it:

> **reference classification / operational target**

and document where it came from.

5.3 Prevent leakage

This is non-negotiable.

If your target is derived from Ibn Hajar's classification:

Ibn Hajar's final classification cannot simultaneously be used as an unrestricted predictor.

Otherwise your model can simply learn the answer.

Separate:

Predictive evidence

- raw assessment text
- critic identity
- graph relationships
- temporal information
- geographical information
- structural graph features

from:

Reference target

- established/reference narrator classification.

PHASE 6 — Build the semantic component

6.1 Extract scholarly assessment text

For each narrator, preserve individual statements rather than only the final grade.

Example:

```
Plain text
Narrator X

Ibn Abi Hatim:
"text..."

al-Dhahabi:
"text..."

Ibn Hajar:
"text..."
```

6.2 Normalise Arabic carefully

Document:

- Unicode normalization
- diacritics
- punctuation
- orthographic variants
- Arabic letters normalization
- tokenisation

Don't destroy information that matters for negation.

For example, the distinction between:

ثقة

and

ليس بثقة

is exactly the type of semantic/contextual information your model should understand.

6.3 Choose the semantic model

Your likely options:

- AraBERT
- Arabic BERT variant
- sentence-transformer style embedding
- fine-tuned transformer

Start with a pretrained model and **only fine-tune if necessary**.

Don't spend FYP2 reinventing Arabic language modelling.

6.4 Decide the semantic representation

Possible:

```
Plain text
assessment text
      ↓
Arabic transformer
      ↓
embedding
      ↓
narrator-level aggregation
```

For multiple critics:

```
critic statements
       ↓
individual embeddings
       ↓
attention / pooling
       ↓
narrator semantic representation
```

Critic identity could be represented separately.

PHASE 7 — Build the graph component

7.1 Construct the graph

At minimum:

```
Narrator → Narrator
```

based on transmission relationships.

Potentially use edge types:

```
reported_from
teacher_of
student_of
```

depending on what your data supports.

7.2 Engineer a simple structural baseline

Before GNN:

- degree
- in-degree
- out-degree
- betweenness
- PageRank
- clustering coefficient
- transmission frequency
- chain position

Then train a conventional model.

This matters because otherwise you can't tell whether the GNN is actually doing anything sophisticated.

7.3 Build the GNN

Possible:

- GCN
- GraphSAGE
- GAT

You don't need all three unless comparing architectures is part of your research.

I'd probably make one the primary architecture and another a lightweight comparison.

PHASE 8 — Build biographical/temporal features

This modality should be intentionally modest.

Examples:

```
death_year
birth_year
tabaqa
teacher_count
student_count
geographic_count
temporal overlap
```

Potentially derive:

> **meeting plausibility**

from temporal overlap.

This is theoretically meaningful and much more interesting than random demographic features.

PHASE 9 — Build the baseline models

This is necessary.

At minimum:

Baseline 0

Simple/reference rule.

Baseline 1

Traditional ML:

- Logistic Regression / Random Forest / XGBoost

depending on the target formulation.

Baseline 2

Semantic-only.

Baseline 3

Graph-only.

Final

Multimodal.

This lets you answer:

> Is the improvement actually coming from the multimodal architecture?

rather than:

> "I trained a GNN and it produced 85%."

PHASE 10 — The core ablation experiment

This is probably the **heart of your FYP2**.

Run:

Model	Semantic	Graph	Temporal/Bio
Baseline	❌	❌	❌
Semantic	✅	❌	❌
Graph	❌	✅	❌
Bio/Temporal	❌	❌	✅
Semantic + Graph	✅	✅	❌
Full	✅	✅	✅

This directly answers your RQs.

And this means your current FYP1 weak graph result becomes useful:

> "Graph-only information was insufficient; the study therefore investigates whether semantic evidence provides complementary information."

PHASE 11 — Evaluate properly

Primary metric

Macro-F1

This should probably be your primary metric if you use multiclass classification.

Why?

Because your classes are likely imbalanced.

Secondary metrics

Use a small, justified set:

- Accuracy
- Weighted-F1
- Cohen's Kappa
- MAE
- Spearman correlation

Depending on whether you're treating the output as categorical, ordinal or continuous.

Don't throw 15 metrics into the thesis just because they're available.

11.1 Confusion matrix

Absolutely include it.

It will show *which reliability categories* the model confuses.

11.2 Error analysis

Take:

- strongest predictions
- weakest predictions
- false positives
- false negatives
- ambiguous identities
- sparse graph nodes

- conflicting scholarly evaluations
- long chains

and explain why they happened.

This will make your discussion much stronger.

PHASE 12 — Solve your chain-level experiment

Once narrator-level estimation works:

```
Plain text
Narrator A → 0.91
Narrator B → 0.87
Narrator C → 0.72
```

you can aggregate them.

But **do not assume product is automatically correct**.

Compare:

Product

$$R_c = \prod_i R_i$$

Minimum

$$R_c = \min(R_i)$$

Mean

$$R_c = \frac{1}{n}\sum R_i$$

Geometric mean

$$R_c = \left(\prod_i R_i\right)^{1/n}$$

Then determine which aggregation behaves best against your reference classification.

This is an excellent small experiment because your current product formulation has the obvious chain-length issue.

PHASE 13 — Validate train/test methodology

This deserves its own checklist.

Check:

- narrator leakage between train/test;
- duplicate chains;
- duplicate texts;
- near-duplicate biographies;
- graph connectivity leakage;
- same narrator appearing in multiple splits;
- target-derived features;
- preprocessing performed before splitting;
- hyperparameter tuning leaking test information.

For a graph task, **randomly splitting edges/nodes without thinking about leakage can seriously inflate results.**

Document your split strategy explicitly.

PHASE 14 — Build explainability

This isn't strictly necessary for the minimum FYP, but I think it would strengthen your final system considerably.

For a narrator:

```
Plain text
```

```
Reliability: 0.78

Semantic evidence:
  positive scholarly assessments
  moderate criticism

Graph evidence:
  well-connected transmission position

Temporal evidence:
  plausible teacher/student overlap
```

The important philosophical distinction:

> **The model explains its computational estimate; it does not claim to explain or override the scholarly tradition itself.**

PHASE 15 — Build the final demonstrator

Your final demo should ideally allow:

```
Plain text
```

```
Enter narrator / isnad
        ↓
Resolve narrator
        ↓
Show profile
        ↓
Show classical evidence
        ↓
Show graph relationships
        ↓
Compute reliability representation
        ↓
Show contributing modalities
        ↓
Optional chain assessment
```

This would be extremely effective during viva.

PHASE 16 — Visualisations

You don't need dozens.

I'd aim for these **seven**:

1. System architecture

Your entire framework.

2. Data pipeline

```
Plain text
```

```
SanadSet → identity resolution → Hadith-Viz → feature extraction → model
```

3. Example narrator profile

Show how heterogeneous evidence becomes structured data.

4. Reliability distribution

Show class/score distribution.

5. Model comparison

Macro-F1 across modalities.

6. Confusion matrix

Final model.

7. Example computational explanation

One narrator / isnād with the contributing evidence.

Also ensure all graphs have readable labels, legends and captions—this directly addresses the FYP1 feedback.

PHASE 17 — Thesis writing

Don't leave this until the end.

Chapter 1 — Introduction

- background
- problem
- research gap
- motivation
- objectives
- research questions
- scope
- significance

Chapter 2 — Literature Review

- ʿIlm al-Rijāl
- jarḥ wa-taʿdīl
- isnād analysis
- hadith computational research
- NLP
- transformers
- knowledge graphs
- GNNs
- related systems
- research gap

End with:

> **What exactly has not been done that your framework addresses?**

Chapter 3 — Methodology

This will be your biggest chapter.

```
Dataset
↓
Data preprocessing
↓
Narrator entity resolution
↓
```

```
Rijāl profile construction
↓
Feature extraction
↓
Graph construction
↓
Model architecture
↓
Experimental design
↓
Evaluation
```

Chapter 4 — Results

Do **not** mix extensive interpretation here.

Present:

- dataset statistics
- resolution accuracy
- baseline performance
- modality performance
- ablation
- chain results
- error analysis

Chapter 5 — Discussion

This is where you answer:

- What does semantic evidence contribute?
- What does graph evidence contribute?
- Why is graph-only weak/strong?
- Why does multimodal integration help/not help?
- What errors remain?
- What are the scholarly implications?
- What are the computational limitations?

Chapter 6 — Conclusion

Answer each RQ explicitly.

Then:

> contributions → limitations → future work.

PHASE 18 — Documentation/reproducibility

Create a proper project structure now.

Something like:

```
fyp2/
├── data/
│   ├── raw/
│   ├── interim/
│   └── processed/
├── notebooks/
├── src/
│   ├── ingestion/
```

```
│   ├── entity_resolution/
│   ├── feature_engineering/
│   ├── semantic/
│   ├── graph/
│   ├── multimodal/
│   └── evaluation/
├── experiments/
├── models/
├── results/
├── visualizations/
└── thesis/
```

Have:

- fixed random seed;
- versioned datasets;
- configuration files;
- experiment logs;
- model checkpoints;
- evaluation scripts;
- SHA-256/data identifiers where appropriate.

Your existing seed-42/data-integrity habits should continue.

PHASE 19 — Final viva preparation

You should be able to answer these **without looking at your slides**:

Why ʿilm al-rijāl?

Why computational quantification?

Why SanadSet?

Why Hadith-Viz?

Why not simply use Ibn Hajar's grade?

Why not use every Hadith-Viz narrator?

How did you resolve narrator identity?

How did you handle homonyms?

How did you prevent data leakage?

Why BERT?

Why GNN?

Why multimodal?

Why Macro-F1?

Why was graph-only performance weak?

What does your numerical reliability score actually mean?

Is your score an objective Islamic judgement?

Answer: no. It is a computational representation derived under an explicitly defined operationalisation.

Does your model replace scholars?

No. It is a computational decision-support framework.

What happens when classical scholars disagree?

The disagreement is retained as evidence rather than silently collapsed into a single "truth."

That last answer is particularly important.

The FYP2 timeline I'd personally use

Weeks 1–2

Freeze research design

RQ → hypotheses → variables → target → metrics → experiment matrix.

Weeks 2–4

Dataset construction

SanadSet → Hadith-Viz → identity resolution → source-complete cohort.

Weeks 4–6

Semantic + biographical features

Get the non-graph side working.

Weeks 5–7

Graph construction + baseline

Rebuild your GNN cleanly.

Weeks 7–9

Core experiments

Baseline → semantic → graph → multimodal → ablation.

Weeks 9–10

Chain-level experiment + error analysis

Weeks 10–11

Explainability + final demonstrator

Weeks 11–13

Results + discussion + thesis

Final weeks

Polish, slides, poster, viva, reproducibility check.

And the most important “DON'T” list

This is probably more valuable than another 20 tasks.

Don't add another major dataset.

No *Tārīkh al-Kabīr* unless it becomes practically free.

Don't try to ingest all 251k Hadith-Viz records.

Use a defensible high-quality cohort.

Don't scrape dozens of additional rijāl books.

Hadith-Viz already gives you the multi-source foundation you need. Its broader corpus currently spans 29 classical works, while the curated/source-supported layer is more appropriate for a controlled experiment.

Don't train five different GNN architectures just because you can.

Don't create an enormous feature zoo.

Don't claim your score is the “true” reliability of a narrator.

Don't claim your model determines ṣaḥīḥ/ḍaʿīf independently of scholarship.

Don't spend months fixing edge-case narrators who aren't needed for the experiment.

Don't wait until the end to write.

Your actual FYP2 in one sentence

I would put this on the wall above your desk:

> **Build a trustworthy narrator cohort from SanadSet + Hadith-Viz, represent classical rijāl evidence and transmission structure as complementary modalities, experimentally compare each modality and their combination, and determine whether the resulting computational representation can meaningfully quantify narrator reliability.**

That's it.

Everything you do should answer **that sentence**.

And honestly, I think this is a much better FYP2 than trying to make the project bigger. **FYP1 discovered the problem; FYP2 should prove the research idea.**